Interpretable Multivariate Conformal Prediction with Fast Transductive Standardization

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Abstract

We propose a conformal prediction method for constructing tight simultaneous prediction intervals for multiple, potentially related, numerical outputs given a single input. This method can be combined with any multi-target regression model and guarantees finite-sample coverage. It is computationally efficient and yields informative prediction intervals even with limited data. The core idea is a novel *coordinate-wise* standardization procedure that makes residuals across output dimensions directly comparable, estimating suitable scaling parameters using the calibration data themselves. This requires neither modeling cross-output dependence nor auxiliary sample splitting. Implementing this idea requires overcoming technical challenges associated with transductive or full conformal prediction. Experiments on simulated and real data demonstrate that this method can produce tighter prediction intervals than existing baselines while maintaining valid simultaneous coverage.

Keywords: interpretability, multivariate analysis, statistical inference, supervised learning, uncertainty quantification.

1 Introduction

1.1 Motivation and Preview of Contributions

We are interested in jointly quantifying uncertainty when predicting several, possibly dependent, continuous outcomes from a shared set of input features. This problem is motivated by many applications where data-driven decisions depend on several unknown variables rather than on any one of them in isolation. For example, a financial institution may use available customer data to jointly forecast that customer's future credit-line utilization, monthly spending volume, and credit score. In this and many other contexts, decision quality hinges on the joint reliability of all predicted values, and an output that is incorrect in even one dimension could lead to adverse consequences.

Conformal prediction is a popular and broadly applicable framework for uncertainty quantification in predictive settings (Vovk et al., 2005; Angelopoulos and Bates, 2023). It derives its main strengths from the ability to augment the output of any machine-learning model with *prediction sets* equipped with finite-sample guarantees, without relying on strong assumptions on the data distribution. Its sole assumption is the availability of labeled data that are *exchangeable* with the test point of interest, a weaker requirement than the standard assumption of i.i.d. sampling. For a *univariate* setting, where the goal is to predict a single continuous-valued outcome, there now exist several well-established conformal prediction methods that produce valid, computationally fast, and easy-to-interpret prediction intervals (Lei et al., 2018; Romano et al., 2019; Sesia and Romano, 2021).

Extending conformal prediction to the *multivariate* setting is a non-trivial problem that has attracted considerable recent interest, spurring a variety of solutions (Dheur et al., 2025). The fundamental challenge is to aggregate a vector of generalized residuals—whose components may differ substantially in scale and variability—into a scalar-valued *non-conformity score* leading to *prediction sets* that should satisfy four key desiderata: (i) finite-sample coverage, (ii) informativeness or tightness, (iii) computational tractability, and (iv) interpretability. Interpretability can take different meanings depending on the application, and in some settings it is consistent with prediction sets having complex shapes. Nevertheless, in many practical scenarios, hyper-rectangular prediction sets—or, equivalently, jointly valid prediction intervals—provide the most natural and transparent way to convey uncertainty to practitioners. This paper focuses on such settings.

Existing methods achieve only a subset of these desiderata. The simplest approach to obtain hyper-rectangular prediction sets is to apply univariate conformal prediction to each output and then take the Cartesian product of the resulting intervals (Neeven and Smirnov, 2018); however, a conservative Bonferroni adjustment to the marginal miscoverage levels is needed to guarantee joint coverage; see Algorithm S1 in Appendix A. This strategy is very easy to implement, but the Bonferroni correction is often too conservative to be practically satisfactory, as it adopts a worst-case view of the dependence structure across output dimensions. By contrast, we aim to achieve valid coverage with joint prediction intervals that are as tight as possible; see Figure 1 for a sketch of this goal.

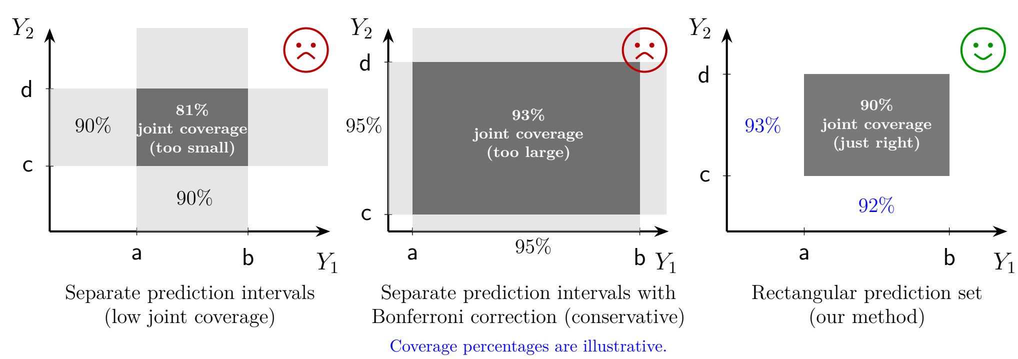

Figure 1: Two-dimensional toy example depicting graphically the main objective of this work: constructing *informative* and *interpretable* hyper-rectangular conformal prediction sets for multivariate outcomes. Left: univariate approaches can produce marginal prediction intervals for each output separately, which do not guarantee joint coverage. Center: a Bonferroni correction can easily restore valid joint coverage, at the cost of being overly conservative. Right: our goal is to efficiently construct hyper-rectangular prediction sets that are easy to interpret and retain valid coverage; doing so involves technical challenges.

A second class of methods reduces multivariate residuals to scalar-valued non-conformity scores, for example by applying an ℓ_2 or ℓ_∞ norm, applies univariate conformal calibration to these scores, and maps the resulting score threshold back into the original output space; see Algorithm S2 in Appendix A. While computationally efficient and able to produce hyper-rectangular prediction sets when using the ℓ_∞ norm, these approaches can be dominated by the noisiest dimension, leading to prediction sets that are unnecessarily wide along low-uncertainty directions.

More sophisticated approaches have been developed to directly minimize the volume of the resulting prediction sets, for example using optimal transport, learned embeddings, copulas, or generative models to describe the dependence structure across residual dimensions. However, these methods aim to solve a problem that is different from that considered in this paper, since they are explicitly designed to produce non-rectangular prediction sets. Moreover, they often introduce substantial computational overhead and require additional independent training data to provide finite-sample guarantees.

In this paper we focus on targeting what we believe is the core difficulty in many applications: the mismatch in the *marginal* scale and variability of residuals across output dimensions. To address this challenge, we develop a method based on transductive coordinate-wise residual standardization. This restores comparability across coordinates and enables the construction of tight hyper-rectangular prediction sets that guarantee finite-sample coverage, without necessitating additional training data or prohibitive computational overhead. Remarkably, our method does *not* need to explicitly model residual dependencies across different dimensions, unlike existing approaches focusing on non-rectangular prediction sets.

Making this idea operational requires overcoming several hurdles, which we resolve through a sequence of increasingly practical steps. We begin by formulating an ideal oracle procedure that assumes access to the test point's true outcome vector and uses this information to standardize the multi-dimensional residuals in the calibration set without violating the exchangeability conditions that are necessary to guarantee finite-sample coverage. Although clearly impractical, this oracle is valuable to guide the design of our practical method. Inspired by the oracle procedure, we introduce two increasingly sophisticated data-driven approaches. The first provides a more conservative approximation of the oracle, whereas the second achieves greater statistical efficiency by building on top of the first. Through extensive numerical experiments, we demonstrate that this progression of ideas culminates in a practical method producing prediction sets satisfying all our desiderata.

1.2 Related Work

Extending conformal prediction to multivariate outcomes has attracted substantial attention in recent years, leading to the development of a wide range of alternative methods.

A prominent line of work seeks minimum-volume prediction sets with flexible geometries by modeling dependence across output dimensions. This includes copula-based approaches (Messoudi et al., 2021; Park and Cho, 2025; Sun and Yu, 2023), non-parametric density estimation methods (Sadinle et al., 2019; Izbicki et al., 2022; Wang et al., 2023; Dheur et al., 2025), parametric density estimation methods (Johnstone and Cox, 2021; Sampson and Chan, 2024; Braun et al., 2025a,b), optimal-transport techniques (Thurin et al., 2025; Klein et al., 2025), and flow-based generative models (Colombo, 2024; Lee et al., 2025; Fang et al., 2025). While these approaches can produce smaller prediction sets than our method, the resulting sets often have complex shapes that are more difficult to interpret. Moreover, these approaches require additional model training on independent data to provide finite-sample guarantees, and many also involve substantial computational overhead.

To the best of our knowledge, the idea of directly addressing heterogeneity across output dimensions via marginal residual rescaling remains relatively under-explored. Baheri and Amiri Shahbazi (2025) pursue a related idea, but rely on a conservative Bonferroni-type correction. The approach of Sampson and Chan (2024) is more closely related to ours, but requires an additional independent dataset; avoiding this requirement is the main technical challenge addressed in our paper. As we show empirically, the method of Baheri and Amiri Shahbazi (2025) is often considerably less efficient than ours—particularly in small-sample regimes—resulting in less stable and less informative prediction sets.

Our solution can be viewed as implementing a special instance of transductive, or full, conformal prediction (Vovk et al., 2005; Vovk, 2013). In contrast to the more commonly used split conformal methods, full conformal prediction repeatedly incorporates the test point, augmented with different candidate labels, directly into a learning algorithm. Although this strategy is computationally prohibitive in general, especially for continuous-valued numerical outcomes and complex algorithms, it can be implemented efficiently in some cases by exploiting structural properties of the underlying predictive model. Accordingly, our paper relates to a broader literature on making full conformal inference computationally practical in special cases, including approaches tailored to sparse generalized linear models (Lei, 2019; Guha et al., 2023) and methods that leverage stability or other forms of structure in the predictive model (Ndiaye and Takeuchi, 2019; Abad et al., 2022; Martinez et al., 2023; Li, 2024). Unlike those works, which focus primarily on accelerating model refitting, we assume a fixed, pre-trained model and instead aim to improve efficiency by avoiding calibration data splitting while explicitly standardizing residuals across output dimensions. Both the problem setting and the proposed methodology are therefore novel.

More broadly, our approach is also related to recent conformal methods that reuse calibration data for different purposes, such as enabling early stopping to control overfitting when training deep learning models (Liang et al., 2023) and leveraging data-adaptive grouping to improve conditional coverage (Zhou and Sesia, 2024; Zhang and Candès, 2024). In contrast to these works, our focus is on addressing heterogeneity across targets in multivariate regression while obtaining interpretable, rectangular prediction sets.

1.3 Outline

This paper is organized as follows. Section 2 introduces notation, a formal problem statement, and the required assumptions, along with a description of an ideal oracle approach. Section 3 presents our methodology, starting with a more conservative approach and then developing more efficient and computationally tractable implementations. Section 4 validates our method empirically and compares its performance to that of existing approaches using both simulated and real data. Section 5 discusses some limitations and possible directions for future research. The appendices contain the remaining mathematical proofs, additional algorithmic details, and further numerical results.

2 Problem Setup and Motivating Constructions

2.1 Notation

For any $n \in \mathbb{N}$, we denote $[n] := \{1, \dots, n\}$. For any $m \in \mathbb{N}$, $\alpha \in (0,1)$, and $(v_1, \dots, v_m) \in \mathbb{R}^m$, let $\widehat{Q}_{1-\alpha}(v_1, \dots, v_m)$ denote the $\lceil (1-\alpha)(m+1) \rceil$-th smallest element of $\{v_1, \dots, v_m, +\infty\}$.

2.2 Setup and Data Assumptions

We consider a multi-output (multivariate) regression setting where the goal is to predict a vector $\mathbf{Y} = (Y_1, \dots, Y_d)$ of d real-valued outputs given input features $\mathbf{X} \in \mathcal{X}$. Assume that $(\mathbf{X}^1, \mathbf{Y}^1), \dots, (\mathbf{X}^{n+1}, \mathbf{Y}^{n+1})$ are jointly exchangeable, or drawn i.i.d. from an arbitrary distribution $P_{X,Y}$. We observe the calibration data $\mathcal{D}_{\text{cal}} = \{(\mathbf{X}^i, \mathbf{Y}^i)\}_{i=1}^n$ and the test covariate $\mathbf{X}^{n+1}$, while the corresponding outcome $\mathbf{Y}^{n+1}$ remains unobserved. Using all of the observed data, the task is to construct a reasonably tight prediction set $\hat{C}(\mathbf{X}^{n+1}) \subset \mathbb{R}^d$ that contains the unknown output $\mathbf{Y}^{n+1}$ with high probability, guaranteeing *marginal coverage*:

$$\mathbb{P}\Big[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1})\Big] \geq 1 - \alpha, \tag{1}$$

where $\alpha \in (0,1)$ is the user-specified miscoverage level. Note that both $\mathbf{Y}^{n+1}$ and $\hat{C}(\mathbf{X}^{n+1})$ are random in (1), as they depend on the new test point as well as on the calibration data $\mathcal{D}_{\text{cal}}$. The term "marginal coverage" means that the guarantee in (1) averages over the joint randomness of the calibration sample and the test point. Stronger notions, such as coverage conditional on each value of the test covariates, are generally unattainable in finite samples without more restrictive assumptions (Foygel Barber et al., 2021). Methods that approximate conditional coverage (Romano et al., 2020; Gibbs et al., 2025) may be combined with our approach. Investigating such combinations is an interesting direction for future work, but is beyond the scope of this paper.

For interpretability, we require the prediction sets to be *hyper-rectangular*, that is, Cartesian products of univariate prediction intervals:

$$\hat{C}(\mathbf{X}^{n+1}) = \bigtimes_{j=1}^{d} \big[L_j(\mathbf{X}^{n+1}), U_j(\mathbf{X}^{n+1})\big], \tag{2}$$

where $[L_j(\mathbf{X}^{n+1}), U_j(\mathbf{X}^{n+1})]$ denotes the coordinate-wise interval for the jth output. Prediction sets in this form decompose naturally across dimensions, making them very easy to explain in the context of multi-output prediction. Consequently, the marginal coverage requirement stated in (1) is equivalent to

$$\mathbb{P}\left[Y_j^{n+1} \in [L_j(\mathbf{X}^{n+1}), U_j(\mathbf{X}^{n+1})] \text{ for all } j \in [d]\right] \geq 1 - \alpha.$$

Thus, although the coverage guarantee is marginal over the calibration and test data, it is simultaneous across all d output coordinates.

2.3 Coordinate-Wise Generalized Residuals

In addition to the data assumptions, we assume access to a *pre-trained* joint predictive model $\hat{\mathbf{f}} = (\hat{f}_1, \dots, \hat{f}_d)$, fitted on a dataset independent of both the calibration set $\mathcal{D}_{\text{cal}}$ and the test point. We treat this model as a black box; the only requirement is that it provide a coordinate-wise *generalized residual* map $\mathbf{E} : \mathcal{X} \times \mathbb{R}^d \to \mathbb{R}^d$, which quantifies the model's prediction error along each output dimension.

Let $\mathbf{E}(\mathbf{X}, \mathbf{Y}) = \big(E_1(\mathbf{X}, Y_1), \ldots, E_d(\mathbf{X}, Y_d)\big) \in \mathbb{R}^d$ denote a vector of coordinate-wise generalized residuals. A classical example for models $\hat{f}_j$ that estimate the conditional mean $\mathbb{E}[Y_j \mid \mathbf{X}]$ (Lei et al., 2018) is $E_j(\mathbf{X}, Y_j) = |Y_j - \hat{f}_j(\mathbf{X})|$. Alternatively, for models that estimate conditional quantiles, as in conformalized quantile regression (Romano et al., 2019), a natural choice is

$$E_j(\mathbf{X}, Y_j) = \max\big\{\hat{f}_j^{\text{lo}}(\mathbf{X}) - Y_j,\ Y_j - \hat{f}_j^{\text{hi}}(\mathbf{X})\big\},$$

where $\hat{f}_j = (\hat{f}_j^{\text{lo}}, \hat{f}_j^{\text{hi}})$ denotes the pair of estimated lower and upper conditional quantiles of $Y_j \mid \mathbf{X}$, for example at levels $\alpha/2$ and $1 - \alpha/2$. Intuitively, this choice of E_j measures the signed distance of Y_j to the nearest boundary of the interval $[\hat{f}_j^{\text{lo}}(\mathbf{X}), \hat{f}_j^{\text{hi}}(\mathbf{X})]$, where positive values indicate that Y_j falls outside the interval, while negative values indicate that it lies inside.

Applying the generalized residual map to the observed calibration data gives $\mathbf{E}^i = \mathbf{E}(\mathbf{X}^i, \mathbf{Y}^i) = (E_1^i, \ldots, E_d^i)$ for all $i \in [n]$. We reserve $\mathbf{E}^{n+1} = \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{Y}^{n+1})$ for the unobserved residual vector of the test point.

Throughout the theoretical development, we assume that the residuals are nonnegative. Residuals with possibly negative coordinates, such as the quantile-based residuals above, can be made nonnegative by truncating them at zero, at the cost of potentially more conservative conformal prediction sets (Romano et al., 2019).

We also impose two additional regularity conditions. First, each residual coordinate is assumed to have finite mean and finite, strictly positive variance at the population level. These moment conditions are mild and automatically satisfied by bounded and nondegenerate residuals. Second, within each coordinate, the calibration and test residuals are assumed to be almost surely distinct, which is a standard assumption in conformal inference. When ties arise, including the ties at zero that may be introduced by truncation, they can be handled through standard randomized tie-breaking or by adding independent continuous jitter to the residuals. These conditions make the theoretical development cleaner, while our numerical experiments indicate that the proposed method remains robust when they are not exactly satisfied. We collect all of the standing conditions below.

Assumption 1 *For every $j \in [d]$, the distribution of the residual $E_j(\mathbf{X}, Y_j)$ satisfies $E_j(\mathbf{X}, Y_j) \geq 0$ almost surely, $\mathbb{E}[E_j(\mathbf{X}, Y_j)] < \infty$, and $0 < \mathrm{Var}[E_j(\mathbf{X}, Y_j)] < \infty$. Moreover, for every $j \in [d]$, the residuals $E_j^1, \ldots, E_j^{n+1}$ are almost surely pairwise distinct:*

$$\mathbb{P}\left(E_j^i \neq E_j^k \text{ for all } 1 \leq i < k \leq n+1\right) = 1.$$

2.4 Conformal Calibration with Marginal Coverage

After evaluating the residuals $\mathbf{E}^i = \mathbf{E}(\mathbf{X}^i, \mathbf{Y}^i)$ for all calibration points indexed by $i \in [n]$, a prediction set for $\mathbf{Y}^{n+1}$ based on $\mathbf{X}^{n+1}$ can be assembled by including all possible values of $\mathbf{y} \in \mathbb{R}^d$ whose residuals $\mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y})$ are sufficiently small:

$$\hat{C}(\mathbf{X}^{n+1}) = \left\{ \mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq W_j, \text{ for all } j \in [d] \right\}, \tag{3}$$

where $W_1, \ldots, W_d$ are suitable (data-dependent) thresholds chosen to satisfy

$$\mathbb{P}\left[0 \leq E_j(\mathbf{X}^{n+1}, Y_j^{n+1}) \leq W_j \text{ for all } j \in [d] \right] \geq 1 - \alpha. \tag{4}$$

This ensures marginal coverage of $\mathbf{Y}^{n+1}$ at level $1-\alpha$, as defined in (1). For standard choices of $\mathbf{E}$, such as absolute residuals, the resulting prediction set $\hat{C}(\mathbf{X}^{n+1})$ is hyper-rectangular, as in (2), with $L_j(\mathbf{X}^{n+1}) = \hat{f}_j(\mathbf{X}^{n+1}) - W_j$ and $U_j(\mathbf{X}^{n+1}) = \hat{f}_j(\mathbf{X}^{n+1}) + W_j$ for all $j \in [d]$. An analogous construction applies to the capped quantile-based residuals described above.

A naive way to achieve (4) is to set $W_j = \hat{Q}_{1-\alpha/d}(E_j^1, \ldots, E_j^n)$, which is equivalent to applying univariate conformal prediction separately across all d outcome dimensions, with a Bonferroni correction (α/d replacing α). This typically leads to very conservative prediction sets because it takes an overly pessimistic worst-case view of possible dependencies of the prediction tasks across outcome dimensions.

We pursue a more efficient approach, beginning with induced *scalar* non-conformity scores $S^i = \Phi(\mathbf{E}^i) \in \mathbb{R}$ for all $i \in [n]$, computed using a scalarization map $\Phi : \mathbb{R}_+^d \to \mathbb{R}$ whose form is discussed below. Then, the prediction set is constructed as:

$$\hat{C}(\mathbf{X}^{n+1}) = \left\{ \mathbf{y} \in \mathbb{R}^d : \Phi(\mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y})) \leq \hat{Q}_{1-\alpha}(S^1, \ldots, S^n) \right\}. \tag{5}$$

If Φ is, in addition, an invertible map, then we can recover the form of (3) from (5) by setting $W_j = \Phi_j^{-1}\left(\hat{Q}_{1-\alpha}(S^1, \ldots, S^n)\right)$. We focus on the case where Φ is invertible, but the coverage guarantee also holds for other choices of Φ, as demonstrated by the following theorem.

Theorem 1 (Generalized Marginal Coverage) *Assume* $(\mathbf{E}^1, \ldots, \mathbf{E}^{n+1})$ *are exchangeable, with* $\mathbf{E}^i = \mathbf{E}(\mathbf{X}^i, \mathbf{Y}^i)$ *for all* $i \in [n+1]$. *Let* $\Phi : \mathbb{R}_+^d \to \mathbb{R}$ *be any* fixed *function, independent of the data. Then,* $\hat{C}(\mathbf{X}^{n+1})$ *defined in* (5) *satisfies* $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1})] \geq 1 - \alpha$. *Moreover, if the scores* $(S^1, \ldots, S^{n+1})$ *given by* $S^i = \Phi(\mathbf{E}^i)$ *are almost surely distinct,* $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1})] \leq 1 - \alpha + \frac{1}{n+1}$. *The same results hold with* Φ *random, if* $(S^1, \ldots, S^{n+1})$ *are exchangeable conditional on it.*

If the induced scalar scores $S^1, \ldots, S^{n+1}$ are almost surely distinct, Theorem 1 guarantees the prediction sets defined in (5) achieve marginal coverage *tightly* from above, as long as the sample size n is not too small. A limitation of Theorem 1, however, is that it cannot tell us whether the coverage is equally tight across different outcome dimensions. In fact, depending on the data distribution, the fitted model, and the choice of Φ, $\hat{C}(\mathbf{X}^{n+1})$ may achieve the desired marginal coverage while being much wider than necessary along certain directions, thereby providing less informative uncertainty estimates (see Figure 6 for an example).

Several recent works propose residual transformations based on ℓ_p-norms, i.e., $\Phi(\cdot) = \|\cdot\|_p$ for different $1 \leq p \leq \infty$ (Johnstone and Ndiaye, 2022). Although conceptually and computationally simple, these can be heavily affected by the noisiest output dimension. A single high-variance dimension may dominate the scalar score and inflate the resulting prediction set uniformly across all coordinates. To obtain tighter prediction sets, recent works have proposed learning a suitable transformation of the residuals prior to applying a norm-based dimension reduction, for example by using optimal transport (Thurin et al., 2025; Klein et al., 2025). However, these approaches require additional training data as well as expensive computations, and they typically lead to prediction sets with irregular shapes.

In this paper, we propose a conceptually simpler solution that consists of *standardizing* the residuals before taking the maximum across coordinates, similar to the ℓ_∞ norm. This leads to rectangular prediction sets with balanced coordinate-wise coverage across all output dimensions, without requiring additional training data or expensive computations. Despite the apparent simplicity of this idea, however, achieving all desiderata while maintaining guaranteed finite-sample coverage involves significant technical challenges, which we overcome as explained below.

2.5 A Population-Standardized Oracle

To motivate our method, we first consider an oracle construction that has access to the coordinate-wise population means and standard deviations of the residuals. Although unavailable in practice, this construction illustrates how coordinate-wise standardization can mitigate heterogeneity in residual scales while preserving validity and rectangular interpretability.

Consider prediction sets of the form in (5), obtained by transforming the vector-valued residuals $\mathbf{E}^i$ using $\Phi : \mathbb{R}^d_+ \to \mathbb{R}$:

$$\Phi(\mathbf{t}; \boldsymbol{\beta}, \boldsymbol{\theta}) := \max_{1 \le j \le d} \frac{t_j - \beta_j}{\theta_j}, \tag{6}$$

where $\boldsymbol{\beta} = (\beta_1, \dots, \beta_d)^\top$ and $\boldsymbol{\theta} = (\theta_1, \dots, \theta_d)^\top$ are vectors of location and scale parameters, respectively. This transformation is invertible, with coordinate-wise inverse $\Phi_j^{-1}(c; \boldsymbol{\beta}, \boldsymbol{\theta}) = c \cdot \theta_j + \beta_j$ for any $c \in \mathbb{R}$. Given these two parameter vectors, we define the level-$(1-\alpha)$ prediction set at $\mathbf{X}^{n+1}$ by

$$\begin{gathered} C(\mathbf{X}^{n+1}; \boldsymbol{\beta}, \boldsymbol{\theta}) = \left\{ \mathbf{y} \in \mathbb{R}^d : 0 \le E_j(\mathbf{X}^{n+1}, y_j) \le \hat{Q}^{\boldsymbol{\beta},\boldsymbol{\theta}}_{1-\alpha} \cdot \theta_j + \beta_j \text{ for all } j \in [d] \right\}, \\ \text{where} \quad \hat{Q}^{\boldsymbol{\beta},\boldsymbol{\theta}}_{1-\alpha} := \hat{Q}_{1-\alpha}(S^1_{\boldsymbol{\beta},\boldsymbol{\theta}}, \dots, S^n_{\boldsymbol{\beta},\boldsymbol{\theta}}) \quad \text{and} \quad S^i_{\boldsymbol{\beta},\boldsymbol{\theta}} = \Phi(\mathbf{E}^i; \boldsymbol{\beta}, \boldsymbol{\theta}). \end{gathered} \tag{7}$$

This set is hyper-rectangular and has the form in (3), with coordinate-wise thresholds $W_j = \Phi_j^{-1}(\hat{Q}^{\boldsymbol{\beta},\boldsymbol{\theta}}_{1-\alpha}; \boldsymbol{\beta}, \boldsymbol{\theta})$. Theorem 1 guarantees that $C(\mathbf{X}^{n+1}; \boldsymbol{\beta}, \boldsymbol{\theta})$ satisfies the marginal coverage requirement in (1) for any fixed choice of $(\boldsymbol{\beta}, \boldsymbol{\theta})$.

To address heterogeneity in scale and variability across dimensions, a natural choice for $(\boldsymbol{\beta}, \boldsymbol{\theta})$ is the pair of coordinate-wise population mean and standard deviation vectors of $\mathbf{E}(\mathbf{X}, \mathbf{Y})$, denoted by $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$:

$$\mu^*_j = \mathbb{E}[E_j(\mathbf{X}, Y_j)] \ge 0, \qquad \sigma^*_j = \sqrt{\mathrm{Var}[E_j(\mathbf{X}, Y_j)]} > 0, \qquad \forall j \in [d]. \tag{8}$$

Then by replacing $(\boldsymbol{\beta}, \boldsymbol{\theta})$ in (7) with the population parameters $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$, we obtain a population-standardized oracle prediction set $C(\mathbf{X}^{n+1}; \boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$.

Although unavailable in practice, this oracle prediction set $C(\mathbf{X}^{n+1}; \boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$ satisfies all four of our desiderata: (i) it guarantees finite-sample marginal coverage, by Theorem 1; (ii) it tends to have balanced coordinate-wise coverage across all output dimensions, because it operates with residuals whose components are all on the same scale; (iii) it is fast to compute, at least for standard choices of the residual function $\mathbf{E}$; (iv) it is an easy-to-interpret hyper-rectangle, at least for standard choices of $\mathbf{E}$.

In the following, we will describe practical approximations of these oracle prediction sets, starting from intuitive but not fully satisfactory solutions before presenting our method.

2.6 Simple but Unsatisfactory Practical Approaches

An intuitive *plug-in* approximation of the ideal oracle is obtained by replacing the population parameters $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$ with their empirical counterparts $(\hat{\boldsymbol{\mu}}^{\text{pi}}, \hat{\boldsymbol{\sigma}}^{\text{pi}})$, evaluated on the observed calibration data, and then plugging in $(\boldsymbol{\beta}, \boldsymbol{\theta}) = (\hat{\boldsymbol{\mu}}^{\text{pi}}, \hat{\boldsymbol{\sigma}}^{\text{pi}})$ into the prediction set formula (7): $\hat{C}^{\text{pi}}(\mathbf{X}^{n+1}) := C(\mathbf{X}^{n+1}; \hat{\boldsymbol{\mu}}^{\text{pi}}, \hat{\boldsymbol{\sigma}}^{\text{pi}})$. This approach, outlined by Algorithm S3 in Appendix A, tends to perform well in large samples (as shown empirically in Appendix C) but lacks finite-sample guarantees. In fact, using a data-dependent transformation Φ, which breaks the exchangeability between calibration and test scores, violates the key assumption of Theorem 1.

An alternative solution that is also intuitive yet unsatisfactory consists of combining the above plug-in approach with an additional data-splitting step that allows one to recover finite-sample guarantees: the location and scale parameter estimates for the transformation function Φ, denoted here as $\hat{\boldsymbol{\mu}}^{\mathbf{ds}}$ and $\hat{\boldsymbol{\sigma}}^{\mathbf{ds}}$, are estimated empirically using only a random subset of $n_1 < n$ observations of the observed calibration data. The remaining $n_2 = n - n_1$ data points are used to evaluate the quantile $\hat{Q}_{1-\alpha}^{\hat{\boldsymbol{\mu}}^{\text{ds}}, \hat{\boldsymbol{\sigma}}^{\text{ds}}}$ and construct $\hat{C}^{\text{ds}}(\mathbf{X}^{n+1}) := C(\mathbf{X}^{n+1}; \hat{\boldsymbol{\mu}}^{\text{ds}}, \hat{\boldsymbol{\sigma}}^{\text{ds}})$. This approach is outlined by Algorithm S4 in Appendix A. While it satisfies the key assumption of Theorem 1, it uses the data inefficiently. Since it reduces the effective number of calibration samples roughly by half, it tends to produce more unstable and often wider-than-necessary prediction sets, especially in small-sample regimes.

The main contribution of this paper is therefore to overcome this inefficiency. We develop a *transductive* conformal prediction method that uses a similar standardization idea but avoids wasteful data splitting by reusing the entire calibration data to estimate the scaling parameters, all while retaining finite-sample guarantees and manageable computational costs. Achieving this requires resolving several technical challenges, as explained step by step below.

3 Methodology

3.1 A Transductive Rescaling Oracle and High-Level Method Blueprint

According to Theorem 1, prediction sets of the form (5) achieve finite-sample marginal coverage if the dimension-reduction map Φ is independent of the calibration data, as assumed in the previous section, or if $(\Phi(\mathbf{E}^1), \ldots, \Phi(\mathbf{E}^{n+1}))$ can remain *exchangeable conditional on* Φ.

This latter requirement, which is strictly weaker than independence, is the foundation of the method developed in this section. We begin by showing how this conditional-exchangeability condition naturally leads to a different *transductive* oracle, which is much more closely aligned with the practical method introduced below.

Consider an imaginary oracle that knows the unordered values in $\{\mathbf{E}^1, \dots, \mathbf{E}^{n+1}\}$. We can estimate the population location and scale parameters $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$ defined in (8) as follows:

$$\hat{\boldsymbol{\mu}}^{\text{oracle}} = (\hat{\mu}_1^{\text{oracle}}, \dots, \hat{\mu}_d^{\text{oracle}})^\top, \quad \hat{\boldsymbol{\sigma}}^{\text{oracle}} = (\hat{\sigma}_1^{\text{oracle}}, \dots, \hat{\sigma}_d^{\text{oracle}})^\top,$$
$$\hat{\mu}_j^{\text{oracle}} = \frac{1}{n+1}\sum_{i=1}^{n+1} E_j^i, \quad \hat{\sigma}_j^{\text{oracle}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n+1}\left(E_j^i - \hat{\mu}_j^{\text{oracle}}\right)^2}. \tag{9}$$

Since these statistics are *symmetric* in all $n+1$ data points and thus invariant to their ordering, conditioning on the random function $\Phi^{\text{oracle}}(\mathbf{t}) := \Phi(\mathbf{t}; \hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$ maintains the exchangeability of $(S^1_{\text{oracle}}, \dots, S^{n+1}_{\text{oracle}})$ where $S^i_{\text{oracle}} := \Phi(\mathbf{E}^i; \hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$. Therefore, Theorem 1 implies that the prediction set $C^{\text{oracle}}(\mathbf{X}^{n+1}) := C(\mathbf{X}^{n+1}; \hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$, obtained by substituting $(\boldsymbol{\beta}, \boldsymbol{\theta}) = (\hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$ into (7), satisfies the marginal coverage requirement in (1). Of course, this oracle is still impractical because both Φ^{oracle} and $\hat{Q}^{\text{oracle}}_{1-\alpha} := \hat{Q}_{1-\alpha}(S^1_{\text{oracle}}, \dots, S^n_{\text{oracle}})$ depend on the unknown test outcome $\mathbf{Y}^{n+1}$ through $\mathbf{E}^{n+1}$.

We will now translate this oracle into a practical method producing a (slightly) larger prediction set that depends only on the observed data. First, we will find suitable *monotonically nondecreasing link functions* $\omega_1, \dots, \omega_d : \mathbb{R} \to \mathbb{R}_+$ satisfying

$$S^{n+1}_{\text{oracle}} \le c \iff 0 \le E_j^{n+1} \le \omega_j(c) \text{ for all } j \in [d], \tag{10}$$

for every $c \in \mathbb{R}$. Using these link functions, we can define an estimated $1-\alpha$ level prediction set of $\mathbf{X}^{n+1}$ by

$$\hat{C}(\mathbf{X}^{n+1}; c) = \left\{\mathbf{y} \in \mathbb{R}^d : 0 \le E_j(\mathbf{X}^{n+1}, y_j) \le \omega_j(c) \text{ for all } j \in [d]\right\}. \tag{11}$$

Substituting $c = \hat{Q}^{\text{oracle}}_{1-\alpha}$ into (10) and (11) gives the equivalence relation

$$\mathbf{Y}^{n+1} \in C^{\text{oracle}}(\mathbf{X}^{n+1}) \iff \mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha}). \tag{12}$$

Although $\hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha})$ is still impractical, it is closer to being computable. The last remaining step is to replace $\hat{Q}^{\text{oracle}}_{1-\alpha}$ with a conservative data-driven estimate, which we denote here as $\hat{Q}^{\text{prac}}_{1-\alpha} \ge \hat{Q}^{\text{oracle}}_{1-\alpha}$. Because $\omega_1(c), \dots, \omega_d(c)$ are nondecreasing in c, doing that will give a practical prediction set

$$\hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{prac}}_{1-\alpha}) \supseteq \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha}), \tag{13}$$

hence inheriting the coverage lower bound from Theorem 1. Technical difficulties, which we address below, arise when one tries to keep this approximation tight and computationally efficient.

3.2 Explicit Form of the Link Functions

The link functions $\omega_1, \dots, \omega_d$ defined implicitly in (10) can be derived by solving the inequalities in (10) for E_j^{n+1} in terms of c, through the observed statistics $\hat{\mu}_j^{\text{obs}}$ and $\hat{\sigma}_j^{\text{obs}}$ defined as

$$\hat{\mu}_j^{\text{obs}} = \frac{1}{n}\sum_{i=1}^{n} E_j^i, \quad \hat{\sigma}_j^{\text{obs}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(E_j^i - \hat{\mu}_j^{\text{obs}}\right)^2}, \quad \forall j \in [d]. \tag{14}$$

The explicit form of $\omega_j(c)$ is given in the following lemma.

Lemma 2 *Under Assumption 1, suitable link functions $\omega_1, \dots, \omega_d$ satisfying* (10) *are:*

$$\omega_j(c) = \begin{cases} 0, & \text{if } c \le -\frac{n}{\sqrt{n+1}}, \\ \max\left\{0,\ \hat{\mu}_j^{\text{obs}} - \hat{\sigma}_j^{\text{obs}}|c|\sqrt{\dfrac{(n+1)^2}{n^2-(n+1)c^2}}\right\}, & \text{if } -\dfrac{n}{\sqrt{n+1}} < c < 0, \\ \hat{\mu}_j^{\text{obs}} + \hat{\sigma}_j^{\text{obs}}|c|\sqrt{\dfrac{(n+1)^2}{n^2-(n+1)c^2}}, & \text{if } 0 \le c < \dfrac{n}{\sqrt{n+1}}, \\ \infty, & \text{if } c \ge \dfrac{n}{\sqrt{n+1}}, \end{cases} \quad \forall j \in [d].$$

Replacing this expression for the link functions in (11) and substituting $c = \hat{Q}_{1-\alpha}^{\text{oracle}}$ give an upper bound $\omega_j(\hat{Q}_{1-\alpha}^{\text{oracle}})$ for $E_j(\mathbf{X}^{n+1}, y_j)$ that is finite (and hence informative) if and only if $\hat{Q}_{1-\alpha}^{\text{oracle}} < n/\sqrt{n+1}$. Fortunately, $\hat{Q}_{1-\alpha}^{\text{oracle}} < n/\sqrt{n+1}$ almost surely as long as the sample size is not too small.

Lemma 3 *Under Assumption 1, if $n \ge 1/\alpha - 1$, then $\hat{Q}_{1-\alpha}^{\text{oracle}} < n/\sqrt{n+1}$ almost surely.*

Armed with Lemma 2 and Lemma 3, what remains to be done is to find a tight upper bound $\hat{Q}_{1-\alpha}^{\text{prac}}$ for $\hat{Q}_{1-\alpha}^{\text{oracle}}$, as previewed in (13). This is the main challenge and focus of the next sections.

3.3 A Conservative Global-Worst-Case Approach

We first introduce a (highly) conservative practical upper bound $\hat{Q}_{1-\alpha}^{\text{gwc}}$, obtained by taking the worst-case upper bound of $\hat{Q}_{1-\alpha}^{\text{oracle}}$ over all possible values of $\mathbf{E}^{n+1} \in \mathbb{R}_+^d$. We refer to the resulting *global-worst-case* (GWC) prediction set as $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) := \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{gwc}})$. As we show in the next section, this conservative prediction set can in fact be sharpened considerably, without sacrificing marginal coverage, albeit with some additional methodological effort.

Let $\mathbf{z} \in \mathbb{R}_+^d$ denote a placeholder for the test residual $\mathbf{E}^{n+1}$. Define the plug-in scaling parameters $\hat{\boldsymbol{\mu}}(\mathbf{z}) := (\hat{\mu}_1(z_1), \dots, \hat{\mu}_d(z_d))$ and $\hat{\boldsymbol{\sigma}}(\mathbf{z}) := (\hat{\sigma}_1(z_1), \dots, \hat{\sigma}_d(z_d))$ as if $\mathbf{E}^{n+1}$ in (9) were replaced by $\mathbf{z}$:

$$\hat{\mu}_j(z_j) = \frac{\sum_{i=1}^{n} E_j^i + z_j}{n+1}, \quad \hat{\sigma}_j(z_j) = \sqrt{\frac{\sum_{i=1}^{n}\left(E_j^i - \hat{\mu}_j(z_j)\right)^2 + \left(z_j - \hat{\mu}_j(z_j)\right)^2}{n}}. \tag{15}$$

With this notation, we have $\hat{\boldsymbol{\mu}}(\mathbf{E}^{n+1}) = \hat{\boldsymbol{\mu}}^{\text{oracle}}$ and $\hat{\boldsymbol{\sigma}}(\mathbf{E}^{n+1}) = \hat{\boldsymbol{\sigma}}^{\text{oracle}}$. Define then the GWC transformation $\hat{\Phi}^{\text{gwc}} : \mathbb{R}_+^d \to \mathbb{R}$ as

$$\hat{\Phi}^{\text{gwc}}(\mathbf{t}) := \max_{1\leq j\leq d} \sup_{z_j \geq 0} \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} \geq \Phi^{\text{oracle}}(\mathbf{t}), \qquad \forall \mathbf{t} \in \mathbb{R}_+^d. \tag{16}$$

Intuitively, for each coordinate $j \in [d]$ and $\mathbf{t} \in \mathbb{R}_+^d$, we consider the most unfavorable (largest) possible value of the standardized ratio as a function of $E_j^{n+1} = z_j \geq 0$. This ensures that $\hat{\Phi}^{\text{gwc}} \geq \Phi^{\text{oracle}}$ uniformly. Conveniently, $\hat{\Phi}^{\text{gwc}}$ admits a closed-form expression.

Lemma 4 *The GWC transformation* $\hat{\Phi}^{\text{gwc}}$ *defined in* (16) *can be written as:*

$$\hat{\Phi}^{\text{gwc}}(\mathbf{t}) = \max_{1\leq j\leq d} \max\left\{ \frac{t_j - \hat{\mu}_j(0)}{\hat{\sigma}_j(0)}, \frac{t_j - \hat{\mu}_j(z_j^*)}{\hat{\sigma}_j(z_j^*)}, -\frac{1}{\sqrt{n+1}} \right\}.$$

where $z_j^* = \hat{\mu}_j^{\text{obs}} - \frac{(\hat{\sigma}_j^{\text{obs}})^2}{t_j - \hat{\mu}_j^{\text{obs}}}$ *if* $\hat{\mu}_j^{\text{obs}} \geq \frac{(\hat{\sigma}_j^{\text{obs}})^2}{t_j - \hat{\mu}_j^{\text{obs}}}$ *and* $z_j^* = 0$ *otherwise.*

Using Lemma 4, we compute the scalar non-conformity scores $S_{\text{gwc}}^i := \hat{\Phi}^{\text{gwc}}(\mathbf{E}^i)$ for $i \in [n]$, and define the corresponding conformal quantile $\hat{Q}_{1-\alpha}^{\text{gwc}} := \hat{Q}_{1-\alpha}(S_{\text{gwc}}^1, \ldots, S_{\text{gwc}}^n)$. Because $\hat{\Phi}^{\text{gwc}} \geq \Phi^{\text{oracle}}$, it follows that $\hat{Q}_{1-\alpha}^{\text{gwc}} \geq \hat{Q}_{1-\alpha}^{\text{oracle}}$ almost surely. Substituting $c = \hat{Q}_{1-\alpha}^{\text{gwc}}$ in (11), we obtain a practical prediction set $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) := \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{gwc}})$ that is guaranteed to cover the unknown test outcome $\mathbf{Y}^{n+1}$ with probability at least $1-\alpha$. Algorithm S5 in Appendix A summarizes this procedure.

Theorem 5 *Assume* $(\mathbf{E}^1, \ldots, \mathbf{E}^{n+1})$ *are exchangeable and Assumption 1 holds. Then,* $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ *constructed by Algorithm S5 satisfies:* $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})] \geq 1-\alpha$.

Proof From (11), it is easy to see that $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \supseteq \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$ almost surely because $\hat{Q}_{1-\alpha}^{\text{gwc}} \geq \hat{Q}_{1-\alpha}^{\text{oracle}}$ and the link functions $\omega_1, \ldots, \omega_d$ outlined in Lemma 2 are monotonically nondecreasing. Then Theorem 1 and the dual relation in (12) together imply that $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})] = \mathbb{P}[\mathbf{Y}^{n+1} \in C^{\text{oracle}}(\mathbf{X}^{n+1})] \geq 1-\alpha$. ■

We refer to Appendix A.3.1 for details on how to construct $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ in practice at cost $\mathcal{O}(dn)$. While computationally efficient, however, $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ is *statistically* inefficient as it tends to be substantially larger than the estimated oracle prediction set $\hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$, which we would ideally like to approximate tightly. In the next section, we therefore extend this approach and develop a tighter *local worst-case* (LWC) prediction set $\hat{C}_{\mathcal{P}}^{\text{lwc}}(\mathbf{X}^{n+1}) \subseteq \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ that approximates the oracle $\hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$ more closely.

3.4 A Tighter Local-Worst-Case Method

The construction of $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ described in the previous section relies on a *global* upper bound $\hat{Q}_{1-\alpha}^{\text{gwc}}$ for the ideal quantile $\hat{Q}_{1-\alpha}^{\text{oracle}}$, designed to hold uniformly over all possible values of the unobserved test residual $\mathbf{E}^{n+1} \in \mathbb{R}_+^d$. However, the resulting prediction sets are often overly conservative, as the GWC procedure treats all realizations of $\mathbf{E}^{n+1}$ equally, even if some choices correspond to $\mathbf{y}$'s that are extremely unlikely to satisfy the condition stated in $\hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha})$ and could therefore be safely discarded.

This motivates the development of a more statistically efficient approach that utilizes a collection of *local* prediction sets, each tailored to a distinct working hypothesis that the true outcome $\mathbf{Y}^{n+1}$ belongs to a specific region of the outcome space. We will first present the high-level overview of this method and then detail the implementation in three steps: (1) we introduce an intuitive, data-driven partition of the outcome space guided by the order statistics of the calibration residuals; (2) we derive a tractable explicit characterization of the local prediction sets; (3) we develop a computationally efficient algorithm for constructing a tight enclosing rectangular prediction set in practice, while avoiding explicit enumeration of an exponential number of local prediction sets.

3.4.1 Method Overview

For any subset $\mathcal{R} \subseteq \mathbb{R}^d$, we consider the working hypothesis that $\mathbf{Y}^{n+1} \in \mathcal{R}$. Let $\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{R})$ denote an upper bound for $\hat{Q}^{\text{oracle}}_{1-\alpha}$ tailored to this hypothesis, which satisfies

$$\mathbf{Y}^{n+1} \in \mathcal{R} \quad \Longrightarrow \quad \hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{R}) \geq \hat{Q}^{\text{oracle}}_{1-\alpha}. \tag{17}$$

Substituting $c = \hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{R})$ in (11), we get a local prediction set

$$\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) := \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{R})) \cap \mathcal{R} \tag{18}$$

that satisfies

$$\mathbf{Y}^{n+1} \in \mathcal{R} \implies \mathcal{R} \supseteq \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \supseteq \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha}) \cap \mathcal{R}. \tag{19}$$

It is worth noting that the containment chain on the right is generally not guaranteed if $\mathbf{Y}^{n+1} \notin \mathcal{R}$. Then the dual relation (12) implies that $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ is guaranteed to cover the unknown test outcome $\mathbf{Y}^{n+1}$ whenever $\mathbf{Y}^{n+1} \in C^{\text{oracle}}(\mathbf{X}^{n+1})$ and $\mathbf{Y}^{n+1} \in \mathcal{R}$ hold simultaneously.

Of course, we do not have the extra information on which fixed region $\mathcal{R}$ contains $\mathbf{Y}^{n+1}$ in practice, but if we can find a sufficiently rich collection of regions $\mathcal{P}$ such that $\mathbf{Y}^{n+1}$ belongs to one of these regions with high probability, we can obtain a valid prediction set by aggregating the local prediction sets $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ over all $\mathcal{R} \in \mathcal{P}$. Furthermore, to ensure interpretability, we report a relatively tight rectangular set that bounds the union of all localized prediction sets, rather than the union itself.

Thanks to our previous GWC construction, we already have a valid candidate for such high-probability regions, namely the global-worst-case prediction set $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$. Let $\mathcal{P}$ be a *random partition* of $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$. In summary, we construct $\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})$ so that it encloses the union of all local prediction sets $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ for $\mathcal{R} \in \mathcal{P}$ with a rectangle, while remaining contained in the global-worst-case prediction set $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$. Letting $\text{Rect}_{\mathcal{P}}(A)$ denote a rectangle that contains the set A while being contained in $\bigsqcup_{\mathcal{R}\in\mathcal{P}} \mathcal{R} = \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$, we define the local-worst-case prediction set as

$$\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1}) := \text{Rect}_{\mathcal{P}}\left(\bigsqcup_{\mathcal{R}\in\mathcal{P}} \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})\right) \subseteq \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}). \tag{20}$$

In practice, our method constructs a *relatively tight*, though not necessarily the *tightest*, such rectangle due to computational considerations. Nonetheless, the resulting prediction sets are (often much) more informative than $\hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$ in finite samples. Moreover, in the large-sample limit, the union in (20) becomes approximately rectangular itself, making this enclosing step a very mild relaxation from the perspective of statistical efficiency.

The following result guarantees that this high-level approach leads to a prediction set $\hat{C}^{\mathrm{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})$ that retains valid marginal coverage.

Proposition 6 *Consider a (random) collection $\mathcal{P}$ of disjoint regions $\mathcal{R} \subseteq \mathbb{R}^d$ satisfying $\sqcup_{\mathcal{R}\in\mathcal{P}}\mathcal{R} = \hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$ almost surely. For any $\mathcal{R} \in \mathcal{P}$, let $\hat{C}^{\mathrm{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \subseteq \mathbb{R}^d$ denote a local prediction set satisfying* (19) *almost surely. Then, $\hat{C}^{\mathrm{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})$ defined in* (20) *satisfies $\hat{C}^{\mathrm{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1}) \subseteq \hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$. Moreover, if $(\mathbf{E}^1, \ldots, \mathbf{E}^{n+1})$ are exchangeable and Assumption 1 holds, $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}^{\mathrm{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})] \geq 1-\alpha$.*

We next introduce an intuitive data-driven partition based on the possible rankings of the test residuals E_j^{n+1} relative to the calibration residuals $(E_j^1, \ldots, E_j^n)$ for each coordinate $j \in [d]$, and then we characterize the corresponding prediction set $\hat{C}^{\mathrm{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1})$.

3.4.2 Step 1: A Data-Driven Partition Based on Sample Order Statistics

For each $j \in [d]$, let $E_j^{(k)}$ denote the k-th order statistic of $\{E_j^i\}_{i=1}^n$, so that $0 := E_j^{(0)} < E_j^{(1)} \leq \ldots \leq E_j^{(n)} < E_j^{(n+1)} := +\infty$. For each multi-index $\mathbf{h} = (h_1, \ldots, h_d) \in [n+1]^d$, define the local region $\mathcal{Y}^{\mathbf{h}}$ in the outcome space as:

$$\begin{aligned} &\mathcal{Y}^{\mathbf{h}} := \left\{ \mathbf{y} \in \mathbb{R}^d : \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y}) \in \mathcal{E}^{\mathbf{h}} \right\}, \quad \text{where} \\ &\mathcal{E}^{\mathbf{h}} := \bigtimes_{j=1}^{d} \left[\, E_j^{(h_j-1)}, U_j^{h_j} \,\right) \quad \text{and} \quad U_j^{h_j} := \min\left\{ E_j^{(h_j)}, \omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha}) \right\}. \end{aligned} \tag{21}$$

Clearly $\mathcal{P}_{n,d} = \{\mathcal{Y}^{\mathbf{h}}\}_{\mathbf{h}\in[n+1]^d}$ partitions $\hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$ into $(n+1)^d$ disjoint regions, some of which may be empty depending on $\omega_1(\hat{Q}^{\mathrm{gwc}}_{1-\alpha}), \ldots, \omega_d(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})$. Figure 2 visualizes this partition and the corresponding construction of $\hat{C}^{\mathrm{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1})$, in a two-dimensional toy example.

While it may appear at first sight that this choice of partition necessarily incurs exponential computational costs of order $\mathcal{O}(n^d)$, rendering it intractable for all but very small dimensions d, in fact it enables a much more efficient characterization of the local prediction sets $\hat{C}^{\mathrm{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^{\mathbf{h}})$ that will lead to a practical algorithm for constructing $\hat{C}^{\mathrm{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1})$ at a computational cost of only $\mathcal{O}(d^2 n \log n)$.

3.4.3 Step 2: Characterization of the Local Prediction Sets

For any $\mathbf{h} \in [n+1]^d$ with $\mathcal{Y}^{\mathbf{h}} \neq \emptyset$, we characterize the local prediction set $\hat{C}^{\mathrm{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^{\mathbf{h}})$ by computing the scalar quantity $\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}})$ that satisfies the constraint (17), following a strategy similar to the worst-case approach from Section 3.3.

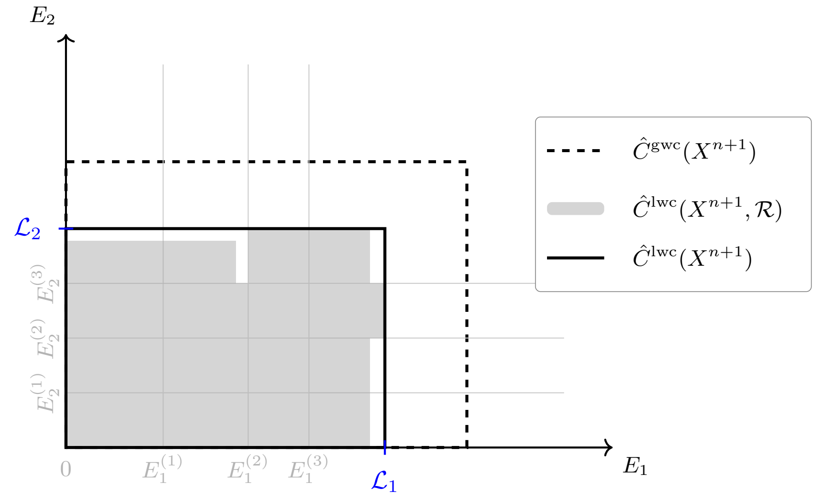

Figure 2: Illustration of the construction of transductively standardized conformal prediction sets using the method outlined in (20), with the data-driven partition $\mathcal{P}_{n,d}$ of the outcome space defined in (21), in a toy example with a two-dimensional outcome variable. The gray grid lines indicate the partition $\mathcal{P}_{n,d}$ in the residual space, as determined by the order statistics $E_j^{(k)}$ of the calibration residuals. The shaded rectangles represent local prediction sets $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ corresponding to different hypothesized rectangles $\mathcal{R} \in \mathcal{P}_{n,d}$. The solid rectangle shows the projection, in residual space, of their rectangular enclosure defined in (20), which constitutes our final prediction set. For comparison, the larger dashed rectangle shows the corresponding projection of the more conservative prediction set $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ from Section 3.3.

Define the LWC scalarization map $\hat{\Phi}^{\text{lwc}}_{\mathbf{h}} : \mathbb{R}^d_+ \to \mathbb{R}$ such that

$$\hat{\Phi}^{\text{lwc}}_{\mathbf{h}}(\mathbf{t}) := \max_{1 \le j \le d} \left\{ \sup_{\mathbf{z} \in \mathcal{E}^{\mathbf{h}}} \frac{t_j}{\hat{\sigma}_j(z_j)} - \inf_{\mathbf{z} \in \mathcal{E}^{\text{gwc}}} \frac{\hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} \right\} \tag{22}$$

where $\mathcal{E}^{\text{gwc}} := \times_{j=1}^{d} \left[0,\, \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})\right]$ is the projection of $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ onto the residual space that satisfies $\mathbf{y} \in \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \iff \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y}) \in \mathcal{E}^{\text{gwc}}$. With this choice, we have

$$\mathbf{E}^{n+1} \in \mathcal{E}^{\mathbf{h}} \implies \hat{\Phi}^{\text{lwc}}_{\mathbf{h}}(\mathbf{t}) \ \ge\ \max_{1 \le j \le d} \sup_{\mathbf{z} \in \mathcal{E}^{\mathbf{h}}} \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} \ge \Phi^{\text{oracle}}(\mathbf{t}), \quad \text{for all } \mathbf{t} \in \mathbb{R}^d_+. \tag{23}$$

By (21), this ensures that for any $\mathbf{h} \in [n+1]^d$ with $\mathbf{Y}^{n+1} \in \mathcal{Y}^{\mathbf{h}} \neq \emptyset$, $\hat{\Phi}^{\text{lwc}}_{\mathbf{h}} \ge \Phi^{\text{oracle}}$ uniformly. Moreover, $\hat{\Phi}^{\text{lwc}}_{\mathbf{h}}(\mathbf{t})$ admits the following closed form.

Lemma 7 *If $\mathcal{Y}^{\mathbf{h}} \neq \emptyset$, then $\hat{\Phi}^{\text{lwc}}_{\mathbf{h}}(\mathbf{t})$ defined in (22) can be written for any $\mathbf{t} \in \mathbb{R}^d_+$ as:*

$$\hat{\Phi}^{\text{lwc}}_{\mathbf{h}}(\mathbf{t}) = \max_{1 \le j \le d} \left\{ \frac{t_j}{r_j(h_j)} - \min\left\{ \frac{\hat{\mu}_j(0)}{\hat{\sigma}_j(0)},\ \lim_{z_j \uparrow \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})} \frac{\hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} \right\} \right\},$$

where $r_j(h_j) = \hat{\sigma}^{\text{obs}}_j$ if $E_j^{(h_j-1)} \le \hat{\mu}^{\text{obs}}_j < U_j^{h_j}$, and otherwise $\min\left\{\hat{\sigma}_j(E_j^{(h_j-1)}), \hat{\sigma}_j(U_j^{h_j})\right\}$.

Note that Lemma 7 involves a limit for $z_j \uparrow \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})$ because $\omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})$ can be infinite when $\hat{Q}^{\text{gwc}}_{1-\alpha} \geq \frac{n}{\sqrt{n+1}}$ (recall Lemma 2), in which case $\mathcal{E}^{\text{gwc}} = \mathbb{R}^d_+$. Nonetheless, our construction remains well-defined even in that special case.

Using Lemma 7, we compute $S^i_{\text{lwc}}(\mathcal{Y}^{\mathbf{h}}) = \hat{\Phi}^{\text{lwc}}_{\mathbf{h}}(\mathbf{E}^i)$ for all $i \in [n]$ and define $\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}}) := \hat{Q}_{1-\alpha}(S^1_{\text{lwc}}(\mathcal{Y}^{\mathbf{h}}), \ldots, S^n_{\text{lwc}}(\mathcal{Y}^{\mathbf{h}}))$. Under the working hypothesis $\mathbf{Y}^{n+1} \in \mathcal{Y}^{\mathbf{h}}$, it follows immediately from (23) that $\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}}) \geq \hat{Q}^{\text{oracle}}_{1-\alpha}$ almost surely. By substituting $c = \hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}})$ in (11), we obtain the local prediction set $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^{\mathbf{h}})$ in (18) with $\mathcal{R} = \mathcal{Y}^{\mathbf{h}}$ and it satisfies (19).

The procedure above summarizes the construction when $\mathcal{Y}^{\mathbf{h}} \neq \emptyset$. If $\mathcal{Y}^{\mathbf{h}} = \emptyset$, we can simply set $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^{\mathbf{h}}) = \emptyset$ and the containment (19) holds trivially. Algorithm 1 summarizes this construction.

Algorithm 1 Construction of local prediction sets under working hypothesis $\mathbf{Y}^{n+1} \in \mathcal{Y}^{\mathbf{h}}$

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; global bounds $\omega_1(\hat{Q}^{\text{gwc}}_{1-\alpha}), \ldots, \omega_d(\hat{Q}^{\text{gwc}}_{1-\alpha})$; calibration mean vector $\hat{\boldsymbol{\mu}}^{\text{obs}}$ and standard deviation vector $\hat{\boldsymbol{\sigma}}^{\text{obs}}$; index $\mathbf{h} \in [n+1]^d$.

1: **if** $\mathcal{Y}^{\mathbf{h}} \neq \emptyset$ **then**
2: Compute $S^i_{\text{lwc}}(\mathcal{Y}^{\mathbf{h}}) \leftarrow \hat{\Phi}^{\text{lwc}}_{\mathbf{h}}(\mathbf{E}^i)$ **for** $i \in [n]$ using $\hat{\Phi}^{\text{lwc}}_{\mathbf{h}}$ defined in Lemma 7.
3: Compute $\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}}) \leftarrow \hat{Q}_{1-\alpha}(S^1_{\text{lwc}}(\mathcal{Y}^{\mathbf{h}}), \ldots, S^n_{\text{lwc}}(\mathcal{Y}^{\mathbf{h}}))$.
4: **return** $\omega_1(\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}})), \ldots, \omega_d(\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}}))$ using ω defined in Lemma 2.
5: **else**
6: **return** $\emptyset$.
7: **end if**

3.4.4 Computational Shortcut for Rectangular Enclosure

While the characterization provided above makes it easy to compute $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^{\mathbf{h}})$ for any specific index $\mathbf{h} \in [n+1]^d$, the problem remains that the intended construction of $\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})$ in (20) involves an *exponential* number of such local prediction sets.

Therefore, to obtain a practical method, we must now develop a shortcut that avoids full enumeration of all $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^{\mathbf{h}})$. Our solution begins by noting that $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^{\mathbf{h}})$ from (18) can be rewritten as:

$$\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^{\mathbf{h}}) = \bigcap_{j=1}^{d} \left\{ \mathbf{y} \in \mathbb{R}^d : E_j^{(h_j-1)} \leq E_j(\mathbf{X}^{n+1}, y_j) \leq B_j^{\mathbf{h}} \right\} \subseteq \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}),$$

where the coordinate-wise upper bounds $B_j^{\mathbf{h}}$ are given by

$$B_j^{\mathbf{h}} := \begin{cases} \min\left\{ U_j^{h_j},\ \omega_j(\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}})) \right\}, & \text{if } U_j^{h_j} > E_j^{(h_j-1)} \text{ and } \omega_j(\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}})) > E_j^{(h_j-1)}, \\ 0, & \text{otherwise.} \end{cases} \tag{24}$$

For every coordinate $j \in [d]$, define the j-th residual boundary $\mathcal{L}_j := \max_{\mathbf{h} \in [n+1]^d} B_j^{\mathbf{h}}$. Then the rectangular enclosure of all local prediction sets in (20) can be expressed as

$$\begin{aligned}\mathrm{Rect}_{\mathcal{P}_{n,d}}\left(\bigsqcup_{\mathbf{h}\in[n+1]^d} \hat{C}^{\mathrm{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^{\mathbf{h}})\right) &= \bigcap_{j=1}^{d} \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \mathcal{L}_j\right\} \\ &=: \hat{C}^{\mathrm{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1}).\end{aligned} \tag{25}$$

Notably, this rectangular enclosure is typically very tight in practice, because the union of the local prediction sets constructed using our designated partition $\mathcal{P}_{n,d}$ is itself very close to being a rectangle when the sample size is large enough.

The last remaining step to make our method practical is finding an efficient algorithm for computing the boundaries $\mathcal{L}_j$ for all $j \in [d]$, without requiring enumeration of all local prediction sets. Such a shortcut is made possible by the following result, which simplifies the definition of $\mathcal{L}_j$ from a maximum over $(n+1)^d$ distinct values to a maximum over an explicit, narrowed-down list of only $n+1$ relevant candidates.

Before stating this result, it is helpful to introduce some notation. Define the (random) *mean index* vector $\mathbf{h}^* \in [n+1]^d$ as the unique index such that $E_j^{(h_j^*-1)} \leq \hat{\mu}_j^{\mathrm{obs}} \leq E_j^{(h_j^*)}$ for all $j \in [d]$. Let $\mathrm{Row}_j(\mathbf{h}^*)$ be a subset of $[n+1]^d$ containing exactly $n+1$ indices, defined as those coinciding with the mean index $\mathbf{h}^*$ on all coordinates other than the j-th one:

$$\mathrm{Row}_j(\mathbf{h}^*) := \left\{\mathbf{h} \in [n{+}1]^d : h_k = h_k^* \text{ for all } k \neq j\right\}, \quad \forall j \in [d].$$

Lemma 8 *If $\mathcal{Y}^{\mathbf{h}^*} \neq \emptyset$, then for any $j \in [d]$, $\mathcal{L}_j := \max_{\mathbf{h}\in[n+1]^d} B_j^{\mathbf{h}}$ can be equivalently written as:*

$$\mathcal{L}_j = \max_{\mathbf{h}\in\mathrm{Row}_j(\mathbf{h}^*)} B_j^{\mathbf{h}} = B_j^{\mathbf{h}^{[j]}}, \quad \textit{where } \mathbf{h}^{[j]} := \operatorname*{argmax}_{\mathbf{h}\in\mathrm{Row}_j(\mathbf{h}^*)} B_j^{\mathbf{h}}.$$

This maximizing index $\mathbf{h}^{[j]}$ satisfies $B_j^{\mathbf{h}} = 0$ for any $\mathbf{h} \in \mathrm{Row}_j(\mathbf{h}^*)$ with $h_j \geq h_j^{[j]}$.

Moreover, for every $j \in [d]$, if there exists $\mathbf{m} \in \mathrm{Row}_j(\mathbf{h}^*)$ with $m_j \geq h_j^*$ and $B_j^{\mathbf{m}} = 0$, then $B_j^{\mathbf{h}} = 0$ for all $\mathbf{h} \in \mathrm{Row}_j(\mathbf{h}^*)$ with $h_j \geq m_j$.

Under the regularity condition $\mathcal{Y}^{\mathbf{h}^*} \neq \emptyset$, which can be immediately verified and typically holds in practice under all but the most extreme scenarios, Lemma 8 gives a very fast algorithm for computing $\mathcal{L}_j$ separately for each coordinate $j \in [d]$. In particular, one can find $\mathbf{h}^{[j]}$, where only $h_j^{[j]}$ needs to be determined, by performing a simple backward search over $\mathbf{h} \in \mathrm{Row}_j(\mathbf{h}^*)$ with $h_j = n+1, \dots, 1$ and stopping at the first h_j with $B_j^{\mathbf{h}} > 0$. Moreover, if $h_j^{[j]} \geq h_j^*$, which can be determined by simply checking whether $B_j^{\mathbf{h}^*} > 0$, we can find $\mathbf{h}^{[j]}$ even more effectively through a binary search that checks whether $B_j^{\mathbf{m}} = 0$ at each intermediate split step $\mathbf{m} \in \mathrm{Row}_j(\mathbf{h}^*)$ with $m_j \in [h_j^*, ..., n+1]$ and assigns $h_j^{[j]}$ to be the largest m_j such that $B_j^{m} > 0$.

The special case $\mathcal{Y}^{\mathbf{h}^*} = \emptyset$ corresponds to an extreme situation where even the conservative GWC prediction set $\hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$ does not contain the empirical mean $\hat{\boldsymbol{\mu}}^{\mathrm{obs}}$ of the calibration residuals. Typically, this should not occur at moderate miscoverage levels α, unless the data are extremely heavy-tailed. Such cases can be handled by simply falling back to the conservative GWC prediction set $\hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$ itself, which we know to have valid marginal coverage. The real-data diagnostics in Appendix C.10.2 report no fallbacks at $\alpha = 0.1$ in the tested splits and examine their frequency at larger miscoverage levels.

In conclusion, the procedure described above, summarized in Algorithm 2, produces:

$$\hat{C}(\mathbf{X}^{n+1}) = \begin{cases} \hat{C}^{\mathrm{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1}), & \text{if } \mathcal{Y}^{\mathbf{h}^*} \neq \emptyset, \\ \hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1}), & \text{otherwise.} \end{cases} \tag{26}$$

Theorem 9 *Assume* $(\mathbf{E}^1, \ldots, \mathbf{E}^{n+1})$ *are exchangeable and Assumption 1 holds. Then,* $\hat{C}(\mathbf{X}^{n+1})$ *constructed by Algorithm 2 satisfies:* $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1})] \geq 1 - \alpha$.

Proof Using the characterization of $\hat{C}(\mathbf{X}^{n+1})$ in (26), the proof follows immediately by combining Proposition 6, which tells us that $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}^{\mathrm{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1})] \geq 1 - \alpha$, and Theorem 5, which tells us that $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})] \geq 1 - \alpha$. ■

Theorem 9 guarantees that our prediction sets $\hat{C}(\mathbf{X}^{n+1})$ defined in (26) have finite-sample marginal coverage of at least $1 - \alpha$. Moreover, these prediction sets have an interpretable rectangular shape and are typically much tighter than $\hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$.

Algorithm 2 Transductively Standardized Conformal Prediction (TSCP)

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0, 1)$; test input $\mathbf{X}^{n+1}$.

Compute $\hat{\boldsymbol{\mu}}^{\mathrm{obs}}, \hat{\boldsymbol{\sigma}}^{\mathrm{obs}}$ using (14).
Compute global bounds $\omega_1(\hat{Q}^{\mathrm{gwc}}_{1-\alpha}), \ldots, \omega_d(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})$ using TSCP-GWC. ▷ Algorithm S5.
Sort $\{E^i_j\}_{i=1}^n$ for each $j \in [d]$ to obtain order statistics $E^{(k)}_j$.
Compute $\mathcal{Y}^{\mathbf{h}^*}$ using (21).
if $\mathcal{Y}^{\mathbf{h}^*} = \emptyset$ **then**
  return $\hat{C}(\mathbf{X}^{n+1}) \leftarrow \left\{ \mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha}), \quad \forall j \in [d] \right\}$.
else
  for $j = 1$ to d **do**
    Compute $\omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*}))$ using Algorithm 1 and $B^{\mathbf{h}^*}_j$ using (24).
    if $B^{\mathbf{h}^*}_j = 0$ **then**
      Backward search over $h_j = h^*_j - 1, \ldots, 1$ to find $\mathbf{h}^{[j]}$. ▷ Algorithm S6
    else
      Binary search over $h_j = h^*_j, \ldots, n{+}1$ to find $\mathbf{h}^{[j]}$. ▷ Algorithm S7
    end if
    Update $\mathcal{L}_j \leftarrow B^{\mathbf{h}^{[j]}}_j$.
  end for
  return $\hat{C}(\mathbf{X}^{n+1}) \leftarrow \left\{ \mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \mathcal{L}_j, \quad \forall j \in [d] \right\}$.
end if

The computational complexity of Algorithm 2 ranges from $\mathcal{O}(d^2 n \log n)$ in the best case, when binary searches can be used for all coordinates, to $\mathcal{O}(d^2 n^2)$ in the worst case, when all coordinates require backward searches; see Appendix A.3 for details. Even in the worst case, a cost of $\mathcal{O}(d^2n^2)$ is typically not prohibitive in practice: calibration sample sizes n for conformal prediction are usually in the hundreds to low thousands, and beyond that additional samples are often better spent improving model training.

In addition, the computation of the distinct boundaries $\mathcal{L}_1, \ldots, \mathcal{L}_d$ is trivially parallelizable, and the entire procedure need not be repeated across different test inputs, since $\mathcal{L}_1, \ldots, \mathcal{L}_d$ are independent of $\mathbf{X}^{n+1}$—remarkably, we have arrived through *transductive* logic at a method that is effectively *inductive*. Therefore, applying Algorithm 2 to m different test points using the same calibration data set has total cost ranging from $\mathcal{O}(d^2 n \log n + dm)$ to $\mathcal{O}(d^2n^2 + dm)$ for typical choices of generalized residuals (Section 2.3).

4 Numerical Experiments

We evaluate Transductively Standardized Conformal Prediction (TSCP) on simulated and real multivariate regression problems, studying joint coverage, coordinate-wise interval length, prediction-set volume, and computational cost. We first consider absolute-residual scores under independent Gaussian noise to establish the effect of unequal outcome scales, followed by dependent and partially heteroskedastic noise to examine cross-output and input-dependent variation. We then study quantile-regression scores and six real datasets. Appendix C provides additional noise-distribution comparisons, sensitivity analyses, heavy-tail and contamination stress tests, and comparisons with the oracle and alternative constructions introduced in the paper.

4.1 Methods and Evaluation Metrics

The primary comparisons use four methods. *Unscaled Max* applies split conformal prediction to the maximum coordinate-wise residual without additional scaling. *Point CHR* estimates coordinate scales using an additional data split (Sampson and Chan, 2024). *Emp. Copula* estimates a dependence model from the calibration residuals and then uses the same observations for calibration; this implementation does not have the finite-sample guarantee enjoyed by the other methods (Messoudi et al., 2021). *TSCP* denotes the rectangular local-worst-case construction proposed in Algorithm 2. For quantile-regression experiments, we replace Point CHR with the native quantile-hyperrectangle method *CQHR* of Sampson and Chan (2024). Appendix C.8 contains a separate low-dimensional comparison with the hyperrectangular shape-template method of Tumu et al. (2024).

For a test sample of size m, we report the empirical joint coverage

$$\widehat{\mathrm{Cov}} = \frac{1}{m}\sum_{i=1}^{m} \mathbf{1}\Big\{\boldsymbol{Y}_i \in \widehat{C}(\boldsymbol{X}_i)\Big\}.$$

For absolute-residual experiments, all methods return a rectangle with residual half-widths $(\mathcal{L}_1, \ldots, \mathcal{L}_d)$. We therefore report the normalized residual-space volume $\prod_{j=1}^{d} \mathcal{L}_j$, which equals 2^{-d} times the full rectangular volume in the outcome space, and the full coordinate-wise interval lengths $2\mathcal{L}_j$. For quantile-regression experiments, the base interval depends on the test covariates, so volume and coordinate lengths are averaged over the test observations in outcome space. Runtime is the wall-clock time needed to construct the conformal region after residuals have been computed; fitting the underlying regression model is excluded. Unless stated otherwise, absolute-residual results are averaged over 200 repetitions (30 repetitions for quantile-regression experiments) and the target coverage is $1 - \alpha = 0.9$. All methods in each repetition use the same fitted regression model, calibration observations, and test observations. Sample sizes and model settings are specified below.

4.2 Absolute Mean-Regression Residuals

We first generate 10-dimensional features $\boldsymbol{X} \sim \mathcal{N}(\mathbf{0}, I_{10})$ and outcomes

$$Y_j = f_j(\boldsymbol{X}) + \epsilon_j, \qquad j \in [d], \tag{27}$$

where $d = 10$ and every f_j is linear, with coefficients sampled once from $\mathrm{Unif}(-10, 10)$ and held fixed across repetitions. In each repetition, we independently generate 7,200 training observations, 800 test observations, and a calibration sample of size $n \in \{30, 50, 100, 300, 500\}$. We estimate the regression coefficients using ordinary least squares, and all methods in each repetition share the same fitted predictor.

4.2.1 Independent Gaussian Noise

Figure 3 reports results under independent Gaussian noise with $\epsilon_j \sim \mathcal{N}(0, (11 - j)^2)$ for $j \in [10]$. TSCP combines near-nominal joint coverage with smaller prediction-set volumes than Point CHR and Unscaled Max. The volume penalty for Point CHR is particularly pronounced when $n < 100$, where its additional split substantially reduces the number of observations available for calibration. Unscaled Max assigns one residual threshold to outputs with unequal noise scales and therefore produces substantially larger rectangles. Emp. Copula yields smaller regions but undercovers; its reuse of calibration observations for dependence estimation precludes the split-conformal guarantee enjoyed by the other benchmarks.

Figure 4 reports the corresponding homogeneous setting, with $\epsilon_j \sim \mathcal{N}(0, 1)$ for all $j \in [10]$. Unscaled Max is the most efficient method with near-nominal coverage, while TSCP remains competitive. Together, these two settings show that coordinate adaptation is useful when noise scales differ, whereas unscaled calibration can be more efficient when a common scale is appropriate.

4.2.2 Heterogeneous and Dependent Noise

The dependent-noise experiment uses $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, DRD)$, where $D = \mathrm{diag}(10, 9, \dots, 1)$ and R is equicorrelated with $\rho = 0.6$. This simultaneously tests heterogeneous coordinate scales and cross-output dependence.

Figure 5 reports joint coverage, residual-space volume, and construction time. As in the independent Gaussian setting with heterogeneous scales, TSCP attains near-nominal coverage with smaller volumes than Point CHR and Unscaled Max.

Figure 6 reports the coordinate-wise interval lengths for all methods at $n = 100$. The mean TSCP full interval length decreases from 49.4 on the noisiest coordinate to 5.0 on the quietest coordinate. In contrast, Unscaled Max assigns the same length, approximately 38.0, to every coordinate. Thus, similar overall coverage does not imply similar coordinate-wise

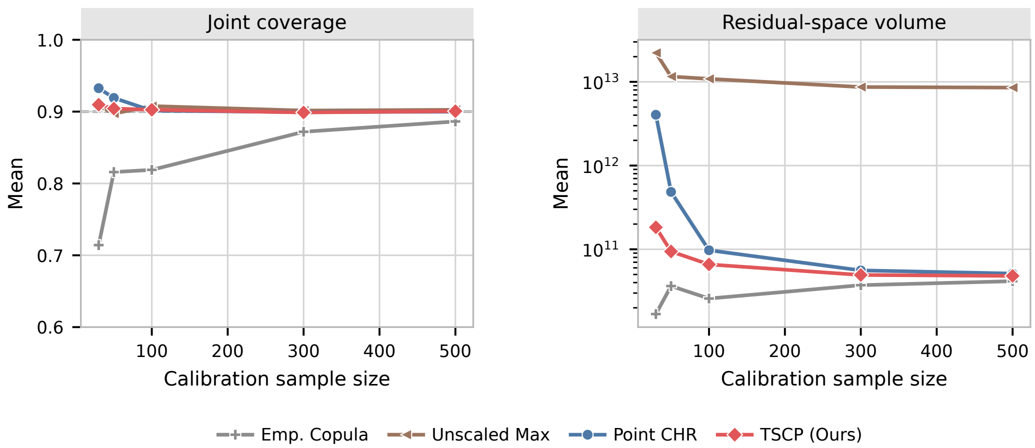

Figure 3: Absolute-residual experiment with $d = 10$, independent Gaussian noise, and heterogeneous scales $(10, 9, \ldots, 1)$. Panels report mean joint coverage and residual-space volume over 200 repetitions. The dashed line marks the target coverage 0.9. TSCP attains near-nominal coverage with smaller residual-space volumes than Point CHR and Unscaled Max.

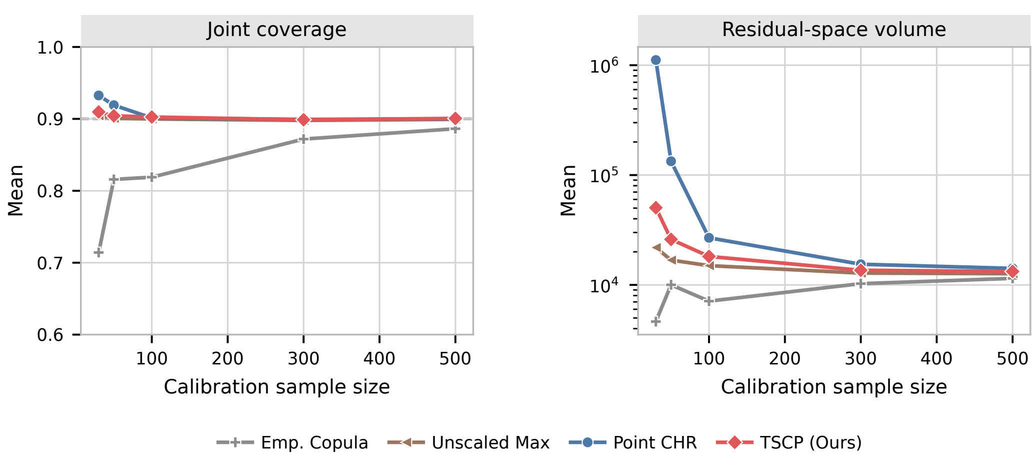

Figure 4: Absolute-residual experiment with $d = 10$, independent Gaussian noise, and unit scale in every coordinate. Panels report mean joint coverage and residual-space volume over 200 repetitions. The dashed line marks the target coverage 0.9. Among methods with near-nominal coverage, Unscaled Max has the smallest residual-space volume, while TSCP remains competitive.

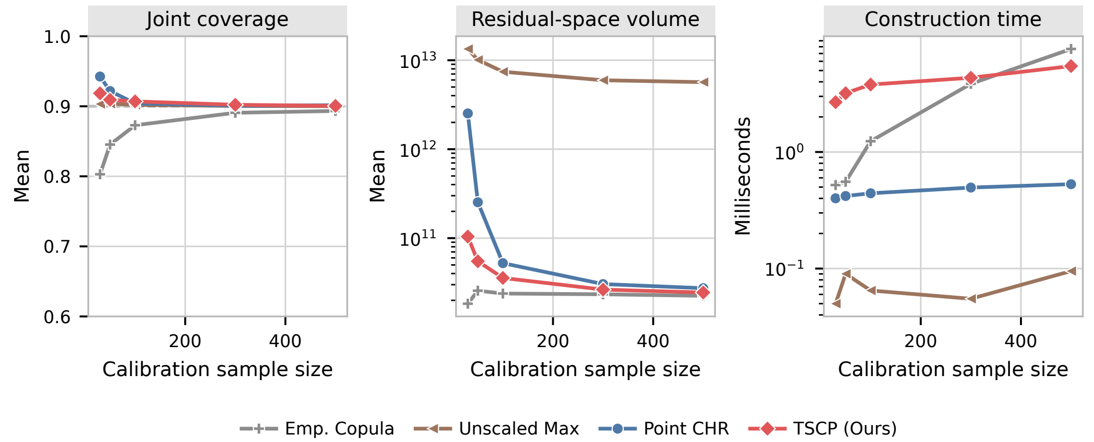

Figure 5: Absolute-residual experiment with $d = 10$, heterogeneous Gaussian scales $(10, 9, \ldots, 1)$, and equicorrelation $\rho = 0.6$. Panels report mean joint coverage, residual-space volume, and construction time over 200 repetitions. The dashed line marks the target coverage 0.9. Runtime excludes model fitting and residual computation.

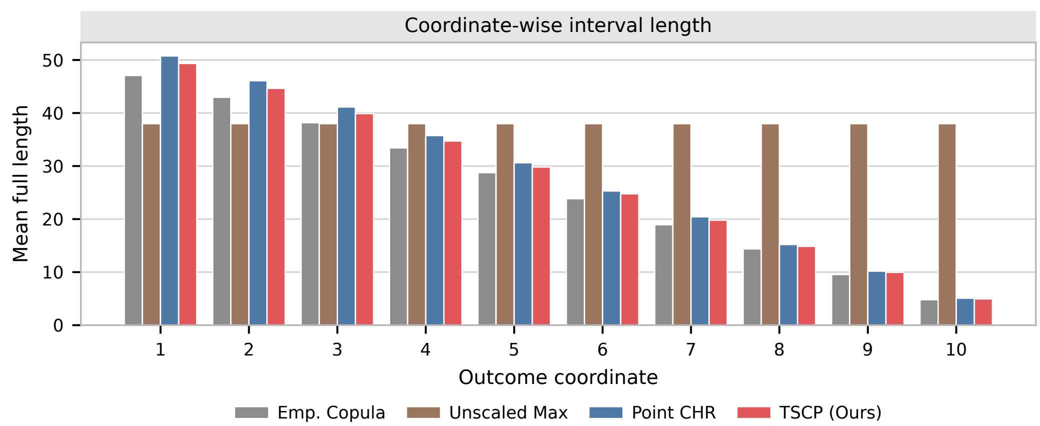

Figure 6: Coordinate-wise mean full interval lengths for the absolute-residual experiment with $d = 10$, $n = 100$, heterogeneous Gaussian scales $(10, 9, \ldots, 1)$, and equicorrelation $\rho = 0.6$. Residual-space half-widths are multiplied by two to report full outcome intervals. Results average 200 independent repetitions. Unscaled Max assigns the same length to every coordinate, while others adapt to the coordinate scales.

informativeness: TSCP allocates width according to coordinate scale instead of allowing the noisiest output to determine all ten intervals. The smaller Emp. Copula bars must be read together with its joint undercoverage in Figure 5.

4.2.3 Partial Heteroskedasticity

To distinguish heterogeneity across coordinates from heteroskedasticity with respect to the input, we next make only the first five coordinates heteroskedastic. For $j \leq 5$, conditional

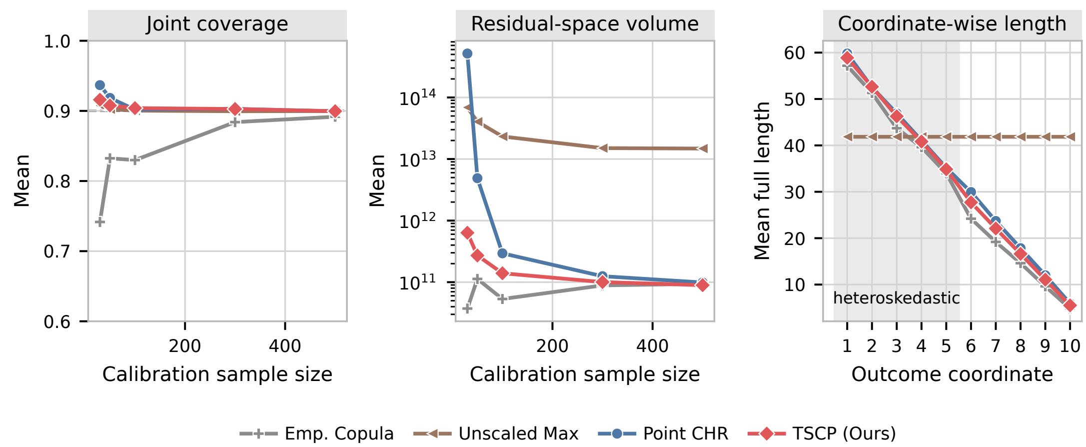

Figure 7: Performance when only the first five of ten outcomes are heteroskedastic in X_1. The shaded coordinates in the right panel are the heteroskedastic ones; full interval lengths are shown for $n = 100$. Results are averaged over 200 repetitions, and the dashed line marks the target joint coverage 0.9.

on $X_1 = x_1$, the noise standard deviation is

$$\sigma_j(x_1) = (11 - j)\sqrt{\frac{1 + 1.5x_1^2}{2.5}},$$

while $\sigma_j(x_1) = 11 - j$ remains constant for $j > 5$. The normalization preserves $\mathbb{E}[\sigma_j^2(X_1)] = (11 - j)^2$, so the comparison isolates input-dependent variance and does not change the marginal second moments. Figure 7 shows that TSCP remains close to the target coverage 0.9 across calibration sizes and continues to adapt coordinate-wise. This experiment does not assert conditional coverage at each value of X_1.

4.3 Quantile-Regression Residuals

We next replace absolute mean-regression residuals with coordinate-wise quantile regression residuals. Let $\widehat{q}_{\ell,j}(x)$ and $\widehat{q}_{u,j}(x)$ denote fitted lower and upper conditional quantiles, ordered pointwise. The signed residual is

$$E_j(x, y) = \max\{\widehat{q}_{\ell,j}(x) - y_j, y_j - \widehat{q}_{u,j}(x)\}.$$

Negative values correspond to observations strictly inside the base interval $[\widehat{q}_{\ell,j}(x), \widehat{q}_{u,j}(x)]$. Because the TSCP theory is stated for nonnegative residuals, we use the capped scores $\max\{E_j, 0\}$. Capping prevents the calibrated interval from shrinking the fitted base interval. We therefore use base-interval miscoverage 0.5 to limit the prevalence of negative signed scores. This choice is not required by the method; Appendix C.7 examines other base-interval levels and a shifted-score transformation. Capping can introduce ties at zero, so these empirical results do not by themselves establish the distinctness condition in Assumption 1.

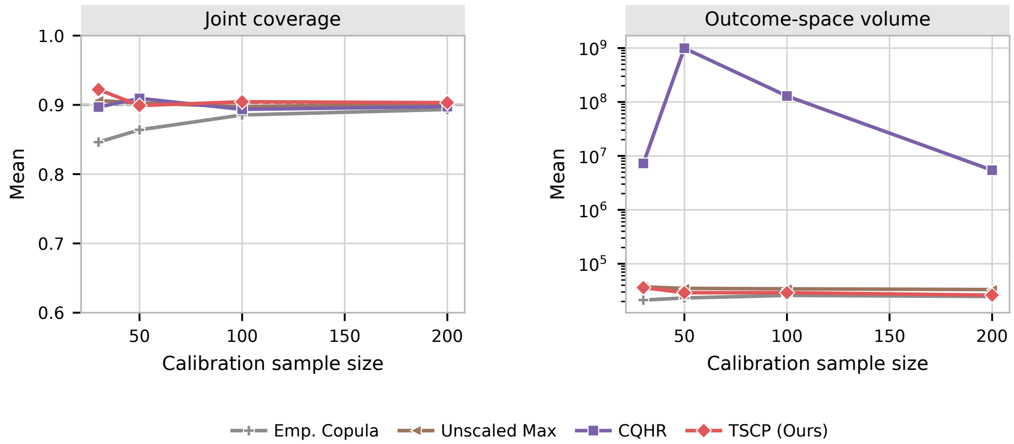

Figure 8: CQR experiment with $d = 3$ and base-interval miscoverage 0.5. TSCP uses capped nonnegative scores; Unscaled Max, CQHR, and Emp. Copula use raw signed scores. Results are averaged over 30 independent repetitions.

Unscaled Max, CQHR, and Emp. Copula use the raw signed residuals for fair comparison. CQHR uses the first coordinate as reference and a numerical floor of 10^{-12} on base-interval lengths when forming width ratios.

We use the same linear signal and ten-dimensional Gaussian features in (27) with $d = 3$ Gaussian outcomes and independent noise scales $(3, 2, 1)$, fit separate gradient-boosted quantile regressions using 2,400 training observations, and evaluate on 600 test observations. Each quantile model uses 100 boosting stages, maximum tree depth 5, learning rate 0.05, and minimum leaf size 5. The base-interval miscoverage is 0.5 and the final target miscoverage is 0.1. Calibration sizes are $n \in \{30, 50, 100, 200\}$. Each of the 30 repetitions uses independently generated training, test, and calibration samples.

Figure 8 shows that TSCP, CQHR, and Unscaled Max attain near-nominal coverage, with TSCP producing the smallest mean outcome-space volume among these three methods in the displayed settings. CQHR's volume is sensitive to the fitted base interval, as examined in Appendix C.7. Emp. Copula undercovers in this comparison.

Figure 9 reports the corresponding coordinate-wise comparison at $n = 100$. In particular, TSCP preserves a substantially shorter third-coordinate interval than Unscaled Max, while CQHR is much wider on the second and third coordinates in this fitted-quantile design. As above, the coordinate-wise lengths should be interpreted together with the joint coverage panel rather than compared without regard to calibration.

4.4 Real-Data Experiments

We evaluate six real-data benchmarks: `rf2` (Tsoumakas et al., 2022), `scm1d` (Contributors, 2019a), `scm20d` (Contributors, 2019b), `stock` (Yeh, 2015), `student` (Cortez, 2008), and `energy` (Tsanas and Xifara, 2012). Each result averages 200 random training/calibration/test splits using proportions 75%/5%/20%. The calibration sizes therefore range from 15 on `stock` to 490 on `scm1d`. A multitask lasso with regularization parameter 10^{-4} is used for `stock`; random forests are used for the other datasets. We use absolute-

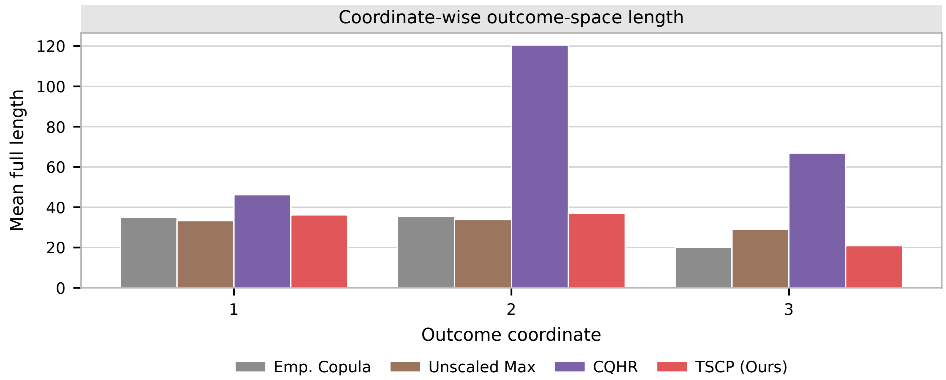

Figure 9: Coordinate-wise mean full outcome-space lengths for the CQR experiment with $d = 3$, $n = 100$, base-interval miscoverage 0.5, and final target coverage 0.9. TSCP uses capped scores, Unscaled Max and Emp. Copula use raw signed scores, and CQHR uses its native construction. Results average 30 independent repetitions.

residual scores and target coverage 0.9 throughout. All methods share the fitted model in each split. We report joint coverage, residual-space volume, and construction time here; coordinate-wise comparisons and search diagnostics appear in Appendix C.10. Table 1 reports coverage and residual-space volume for the four primary methods.

Among TSCP, Point CHR, and Unscaled Max, TSCP has the smallest average volume on `stock`, `scm1d`, `scm20d`, and `energy`. Point CHR has the smallest volume on `rf2`, while Unscaled Max has the smallest volume on `student`. Thus, the coordinate-adaptive construction is beneficial in several datasets but is not uniformly preferable. Emp. Copula produces smaller regions with empirical coverage below the target.

Figure 10 compares construction times. TSCP constructs a region in 0.74–7.46 milliseconds on these datasets. It is slower than the algebraically simpler Unscaled Max and Point CHR implementations, but substantially faster than Emp. Copula on the two 16-dimensional SCM datasets.

The infinite Point CHR volume on `stock` should not be interpreted as an intrinsic failure of the method. With only $n = 15$ calibration observations, its additional split leaves too few observations for a finite 0.9 conformal quantile. A practitioner committed to Point CHR would allocate a larger fraction of the data to its final calibration stage, at the cost of fewer observations for model and scale estimation.

4.4.1 Interpretability of Rectangular Prediction Sets

To emphasize their interpretability, Table 2 displays the rectangular prediction sets produced by different methods for two representative test points from the *energy* dataset, where $d = 2$. In this setting, rectangular prediction sets consist of pairs of simultaneous prediction intervals that aim to jointly contain the true outcomes with probability 90%.

Dataset	Method	d	Coverage	Volume
`stock`	Unscaled Max	6	0.940 (0.057)	6.03e−2 (1.57e−1)
	Emp. Copula	6	0.727 (0.113)	1.01e−5 (9.80e−6)
	Point CHR	6	1.000 (0.000)	∞
	TSCP	6	0.956 (0.059)	**1.86e−3 (8.55e−3)**
`rf2`	Unscaled Max	8	0.900 (0.016)	4.14e3 (2.90e3)
	Emp. Copula	8	0.893 (0.016)	5.35e1 (4.69e1)
	Point CHR	8	0.901 (0.022)	**6.85e1 (7.55e1)**
	TSCP	8	0.901 (0.017)	5.42e2 (1.50e3)
`scm1d`	Unscaled Max	16	0.900 (0.016)	2.74e39 (4.02e39)
	Emp. Copula	16	0.893 (0.016)	1.09e39 (1.34e39)
	Point CHR	16	0.905 (0.020)	2.84e39 (5.43e39)
	TSCP	16	0.901 (0.015)	**1.52e39 (2.61e39)**
`scm20d`	Unscaled Max	16	0.901 (0.016)	2.70e40 (2.92e40)
	Emp. Copula	16	0.892 (0.017)	1.18e40 (1.19e40)
	Point CHR	16	0.902 (0.023)	3.00e40 (8.00e40)
	TSCP	16	0.901 (0.016)	**1.53e40 (1.50e40)**
`energy`	Unscaled Max	2	0.917 (0.049)	1.58e1 (6.80e0)
	Emp. Copula	2	0.886 (0.061)	5.41e0 (4.49e0)
	Point CHR	2	0.899 (0.069)	8.81e0 (2.42e1)
	TSCP	2	0.922 (0.048)	**6.95e0 (4.43e0)**
`student`	Unscaled Max	3	0.904 (0.055)	**1.51e2 (1.27e2)**
	Emp. Copula	3	0.861 (0.073)	1.08e2 (7.19e1)
	Point CHR	3	0.939 (0.057)	6.89e2 (1.07e3)
	TSCP	3	0.912 (0.056)	1.98e2 (1.77e2)

Table 1: Performance on real datasets, averaged over 200 random splits. One standard deviation is reported in parentheses. Bold volume denotes the smallest average volume among TSCP, Point CHR, and Unscaled Max. Red coverage values are more than two Monte Carlo standard errors below 0.9, where the standard error is the across-split standard deviation divided by $\sqrt{200}$. This is a pointwise descriptive flag, not a simultaneous test or a proof of validity. The calibration sizes are 15, 384, 490, 448, 38, and 32 in the order shown.

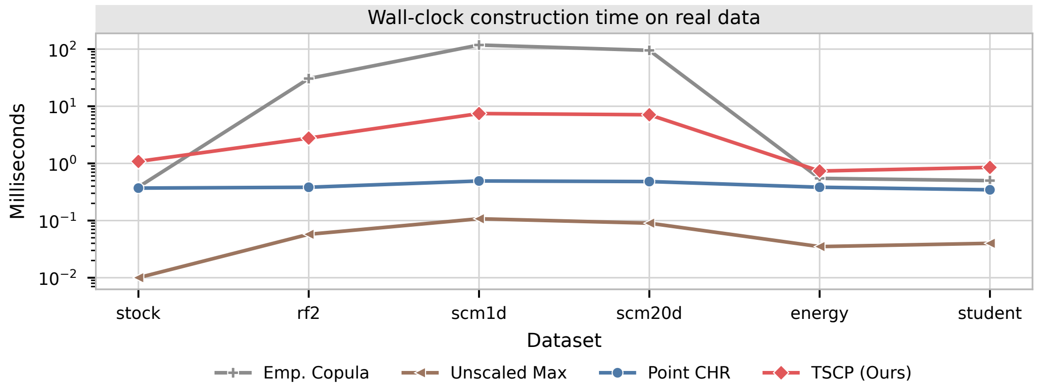

Figure 10: Average wall-clock time for conformal region construction on the six real datasets. The vertical axis is logarithmic and reports milliseconds. All methods use the same fitted regression model in each split; model fitting and residual computation are excluded.

Test 1/Methods	Outcome 1	Outcome 2	Volume	Ground Truth
Unscaled Max	[31.70, 38.94]	**[34.74, 41.98]**	13.09	(35.40, 39.22)
Emp. Copula	**[34.36, 36.28]**	**[34.74, 41.98]**	3.47	
Point CHR	[34.29, 36.36]	[31.45, 45.27]	7.14	
TSCP	[34.28, 36.36]	[33.63, 43.08]	4.93	
Test 2/Methods	Outcome 1	Outcome 2	Volume	Ground Truth
Unscaled Max	[25.44, 32.67]	[29.34, 36.58]	13.09	(29.88, 28.31)
Emp. Copula	**[28.10, 30.02]**	[29.34, 36.58]	3.47	
Point CHR	[28.02, 30.09]	[26.05, 39.87]	7.14	
TSCP	[28.01, 30.10]	**[28.24, 37.69]**	4.93	

Table 2: Illustration of rectangular conformal prediction sets for two test points from the *energy* dataset. Volumes are computed in residual space. Intervals that fail to cover the corresponding true outcomes are highlighted in red; the shortest interval containing each true outcome is highlighted in **bold**. This example is illustrative: conformal coverage guarantees hold on average, not for individual test points.

5 Discussion and Future Work

The core idea underlying the method presented in this paper is the explicit standardization of residuals, which contributes both to the transparency and to the computational efficiency of the resulting prediction sets. As with other standardization-based methods, however, this strategy introduces some sensitivity to outliers. This behavior is visible in the numerical experiments based on the *rf2* dataset, where a small number of influential observations can substantially expand the resulting prediction sets. The contamination study in Appendix C.6 likewise distinguishes stable empirical coverage from substantial volume inflation. An important direction for future work is therefore the development of robust extensions of our method, for example through rescaling based on median absolute deviation or quantile-based normalizations, while preserving the exchangeability required for conformal validity.

Another direction is a direct treatment of signed residuals without capping or an additive shift. The CQR comparisons in Section 4.3 and Appendix C.7 illustrate the limitations of these transformations. A direct extension could enable shrinkage of baseline prediction sets when starting from quantile regression models that already exhibit conservative coverage. It would also be of interest to adapt the proposed approach to settings with non-numerical outcomes, such as categorical or ordinal responses. Finally, our method could be extended to handle partly observed outcomes (Braun et al., 2025b).

Software Availability

A software implementation of the methods described in this paper, along with code necessary to reproduce the numerical experiments, is available online at `https://github.com/OTheMagic/multi-target-scaling.git`.

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Appendix: Algorithms, Proofs, and Numerical Results

Appendix A. Additional Algorithmic Details

A.1 Implementation of Alternative Methods

Algorithm S1 Separate Prediction Intervals with Bonferroni Correction

---

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

1: Compute $\alpha^* \leftarrow \alpha/d$.
2: **for** $j = 1$ to d **do**
3: Compute $\hat{Q}_{1-\alpha^*,j} \leftarrow \hat{Q}_{1-\alpha^*}(E_j^1, \ldots, E_j^n)$.
4: **end for**
5: **return** $\hat{C}^{\text{Bonf.}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}_{1-\alpha^*,j}, \forall j \in [d]\right\}$.

Algorithm S2 Prediction Intervals via Dimension Reduction with ℓ_∞-Norm (Unscaled)

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

1: Compute $S^i_{\text{unscaled}} \leftarrow \|\mathbf{E}^i\|_\infty$ for $i = 1, \ldots, n$.
2: Compute $\hat{Q}^{\text{unscaled}}_{1-\alpha} \leftarrow \hat{Q}_{1-\alpha}(S^1_{\text{unscaled}}, \ldots, S^n_{\text{unscaled}})$
3: **return** $\hat{C}^{\text{unscaled}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}^{\text{unscaled}}_{1-\alpha}, \forall j \in [d]\right\}$.

Algorithm S3 Heuristic Plug-In Standardization

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

1: **for** $j = 1$ to d **do**
2: Compute the location and scale parameters

$$\hat{\mu}_j^{\text{pi}} \leftarrow \frac{1}{n}\sum_{i=1}^n E_j^i, \qquad \hat{\sigma}_j^{\text{pi}} \leftarrow \sqrt{\frac{1}{n-1}\sum_{i=1}^n (E_j^i - \hat{\mu}_j^{\text{pi}})^2}.$$

3: **end for**
4: Compute $S^i_{\text{pi}} \leftarrow \Phi(\mathbf{E}^i; \hat{\boldsymbol{\mu}}^{\text{pi}}, \hat{\boldsymbol{\sigma}}^{\text{pi}})$ for $i \in [n]$, using Φ defined in (6).
5: Compute $\hat{Q}^{\text{pi}}_{1-\alpha} \leftarrow \hat{Q}_{1-\alpha}(S^1_{\text{pi}}, \ldots, S^n_{\text{pi}})$.
6: Compute

$$\hat{C}^{\text{pi}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}^{\text{pi}}_{1-\alpha} \cdot \hat{\sigma}_j^{\text{pi}} + \hat{\mu}_j^{\text{pi}}, \forall j \in [d]\right\},$$

which is hyper-rectangular for the standard choices of $\mathbf{E}$ discussed in Section 2.4.
7: **return** Prediction set $\hat{C}^{\text{pi}}(\mathbf{X}^{n+1})$.

Algorithm S4 Standardization with Data Splitting

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.
Output: $\hat{C}^{\text{ds}}(\mathbf{X}^{n+1})$

1: Randomly split $\{1,\dots,n\}$ into two disjoint subsets, $\mathcal{I}_1$ and $\mathcal{I}_2$.
2: **for** $j = 1$ to d **do**
3: Compute the location and scale parameters

$$\hat{\mu}_j^{\text{ds}} \leftarrow \frac{1}{|\mathcal{I}_1|}\sum_{i\in\mathcal{I}_1} E_j^i, \qquad \hat{\sigma}_j^{\text{ds}} \leftarrow \sqrt{\frac{1}{|\mathcal{I}_1|-1}\sum_{i\in\mathcal{I}_1}(E_j^i - \hat{\mu}_j^{\text{ds}})^2}.$$

4: **end for**
5: Compute $S_{\text{ds}}^i \leftarrow \Phi(\mathbf{E}^i; \hat{\boldsymbol{\mu}}^{\text{ds}}, \hat{\boldsymbol{\sigma}}^{\text{ds}})$ for $i \in \mathcal{I}_2$, using Φ defined in (6).
6: Compute $\hat{Q}_{1-\alpha}^{\text{ds}} \leftarrow \hat{Q}_{1-\alpha}(S_{\text{ds}}^1,\dots,S_{\text{ds}}^n)$.
7: **return** $\hat{C}^{\text{ds}}(\mathbf{X}^{n+1}) \leftarrow \Big\{\mathbf{y}\in\mathbb{R}^d : 0 \le E_j(\mathbf{X}^{n+1}, y_j) \le \hat{Q}_{1-\alpha}^{\text{ds}}\cdot\hat{\sigma}_j^{\text{ds}} + \hat{\mu}_j^{\text{ds}}, \forall j\in[d]\Big\}$.

A.2 Auxiliary Algorithms

Algorithm S5 Global-Worst-Case TSCP (TSCP-GWC)

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

1: Compute $\hat{\boldsymbol{\mu}}^{\text{obs}}, \hat{\boldsymbol{\sigma}}^{\text{obs}}$ using (14).
2: Compute $S_{\text{gwc}}^i \leftarrow \hat{\Phi}^{\text{gwc}}(\mathbf{E}^i)$ using $\hat{\Phi}^{\text{gwc}}$ defined in Lemma 4.
3: Compute $\hat{Q}_{1-\alpha}^{\text{gwc}} \leftarrow \hat{Q}_{1-\alpha}(S_{\text{gwc}}^1,\dots,S_{\text{gwc}}^n)$.
4: Compute $\omega_1(\hat{Q}_{1-\alpha}^{\text{gwc}}),\dots,\omega_d(\hat{Q}_{1-\alpha}^{\text{gwc}})$ using ω defined in Lemma 2.
5: **return** $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \leftarrow \Big\{\mathbf{y}\in\mathbb{R}^d : 0 \le E_j(\mathbf{X}^{n+1}, y_j) \le \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}}), \forall j\in[d]\Big\}$.

Algorithm S6 Backward Search in Algorithm 2

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; global bounds $\omega_1(\hat{Q}_{1-\alpha}^{\text{gwc}}),\dots,\omega_d(\hat{Q}_{1-\alpha}^{\text{gwc}})$; target miscoverage level $\alpha \in (0,1)$; calibration mean $\hat{\boldsymbol{\mu}}^{\text{obs}}$ and standard deviation $\hat{\boldsymbol{\sigma}}^{\text{obs}}$; mean index $\mathbf{h}^*$; current coordinate j.

1: **for** $\mathbf{h} \in \text{Row}_j(\mathbf{h}^*)$ with $h_j = h_j^* - 1$ to 1 **do**
2: Compute $\omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{\mathbf{h}}))$ using Algorithm 1 and $B_j^{\mathbf{h}}$ using (24).
3: **if** $B_j^{\mathbf{h}} > 0$ **then**
4: **return** $\mathbf{h}^{[j]} = \mathbf{h}$.
5: **end if**
6: **end for**

Algorithm S7 Binary Search in Algorithm 2

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; global bounds $\omega_1(\hat{Q}^{\text{gwc}}_{1-\alpha}), \dots, \omega_d(\hat{Q}^{\text{gwc}}_{1-\alpha})$; target miscoverage level $\alpha \in (0,1)$; calibration mean $\hat{\boldsymbol{\mu}}^{\text{obs}}$ and standard deviation $\hat{\boldsymbol{\sigma}}^{\text{obs}}$; mean index $\mathbf{h}^*$; current coordinate j.

```
Let I_min ← h*_j and I_max ← n + 1.
while I_min < I_max do
    Compute m_j ← ⌈(I_min + I_max)/2⌉ and m_k = h*_k for all k ≠ j.
    Compute ω_j(Q̂^lwc_{1−α}(𝒴^m)) and B_j^m using (24).        ▷ Algorithm 1
    if B_j^m > 0 then
        I_min ← m_j.
    else
        I_max ← m_j − 1.
    end if
    if I_max = I_min then
        return h_j^[j] = I_min and h_k^[j] = h*_k for all k ≠ j.
    end if
end while
```

A.3 Computational Cost Analysis

A.3.1 Algorithm S5

Since the calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$ are stored as an (n,d)-array, we know

1. Computing $(\hat{\mu}_j^{\text{obs}}, \hat{\sigma}_j^{\text{obs}})_{j=1}^d$ using (14) requires $\mathcal{O}(nd)$.
2. Evaluating $S^i_{\text{gwc}} \leftarrow \hat{\Phi}^{\text{gwc}}(\mathbf{E}^i)$ for all $i \in [n]$ is also $\mathcal{O}(nd)$.
3. Computing the quantile $\hat{Q}^{\text{gwc}}_{1-\alpha}$ costs $\mathcal{O}(n)$ using a selection algorithm such as quickselect.
4. Computing $\omega_1(\hat{Q}^{\text{gwc}}_{1-\alpha}), \dots, \omega_d(\hat{Q}^{\text{gwc}}_{1-\alpha})$ costs $\mathcal{O}(d)$ once $\hat{Q}^{\text{gwc}}_{1-\alpha}$ and $(\hat{\mu}_j^{\text{obs}}, \hat{\sigma}_j^{\text{obs}})_{j=1}^d$ are known.

Hence, the overall computational complexity of Algorithm S5 is $\mathcal{O}(nd)$.

A.3.2 Algorithm 1

Algorithm 1 shares the same structure and computational cost as Algorithm S5: $\mathcal{O}(nd)$.

A.3.3 Algorithm 2

Since the calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$ are stored as an (n,d)-array, we know

1. Computing $(\hat{\mu}_j^{\text{obs}}, \hat{\sigma}_j^{\text{obs}})_{j=1}^d$ using (14) and $\omega_1(\hat{Q}^{\text{gwc}}_{1-\alpha}), \dots, \omega_d(\hat{Q}^{\text{gwc}}_{1-\alpha})$ using Algorithm S5 requires $\mathcal{O}(nd)$.
2. Sorting $\{E_j^i\}_{i=1}^n$ for all $j \in [d]$ costs $\mathcal{O}(dn \log n)$;

3. Computing $\mathcal{Y}^{\mathbf{h}^*}$ relies on computing $\mathbf{h}^*$, which requires masking all values below $\hat{\boldsymbol{\mu}}^{\text{obs}}$ and reporting the index of the lowest value from the sorted residuals list $\{E_j^i\}_{i=1}^n$, each of which costs $\mathcal{O}(n)$, so together this operation contributes $\mathcal{O}(dn)$.

4. If $\mathcal{Y}^{\mathbf{h}^*} = \emptyset$, then the cost is only $\mathcal{O}(d)$ since all global bounds have been computed.

5. If $\mathcal{Y}^{\mathbf{h}^*} \neq \emptyset$, then for each coordinate, we compute the local bounds using Algorithm 1 and $B_j^{\mathbf{h}^*}$ using (24), which costs $\mathcal{O}(dn)$. If $B_j^{\mathbf{h}^*} = 0$, then we use the backward search detailed in Algorithm S6, which costs $\mathcal{O}(dn^2)$; see Appendix A.3.4. If $B_j^{\mathbf{h}^*} > 0$, then we use the binary search detailed in Algorithm S7, which costs $\mathcal{O}(dn \log n)$; see Appendix A.3.5.

Let $T \leq d$ be the number of coordinates where binary searches are enabled. Then the total cost of Algorithm 2 is $\mathcal{O}(Tdn \log n + (d - T)dn^2)$.

A.3.4 Algorithm S6

Running Algorithm 1 and forming $B_j^{\mathbf{h}}$ costs $\mathcal{O}(dn)$ for each $\mathbf{h}$; there are $n + 1$ indices in $\text{Row}_j(\mathbf{h}^*)$, so the computational complexity of Algorithm S6 never exceeds $\mathcal{O}(dn^2)$.

A.3.5 Algorithm S7

Running Algorithm 1 and forming $B_j^{\mathbf{h}}$ costs $\mathcal{O}(dn)$ for each $\mathbf{h}$. The number of indices $\mathbf{h}$ encountered in this algorithm is $\mathcal{O}(\log_2 n)$ because each step retains only half of the current index range at each step, as in a standard binary search. Therefore, the total computational complexity of Algorithm S7 never exceeds $\mathcal{O}(dn \log n)$.

Appendix B. Mathematical Proofs

B.1 Auxiliary Lemmas

Lemma S1 *Let $x^1, \ldots, x^n \in \mathbb{R}$ and $\bar{x}^n = \frac{1}{n}\sum_{i=1}^n x^i$. For any $x^{n+1} \in \mathbb{R}$, let $\hat{\sigma}(x^{n+1})$ be the sample standard deviation of $\{x^1, \ldots, x^n, x^{n+1}\}$. Then*

$$\hat{\sigma}^2(x^{n+1}) = \frac{(x^{n+1} - \bar{x}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2.$$

Proof Denote $\bar{x}^{n+1} = \frac{1}{n+1}\sum_{i=1}^{n+1} x^i$. We notice that

$$\begin{aligned}
\hat{\sigma}^2(x^{n+1}) =& \frac{1}{n}\sum_{i=1}^{n+1}(x^i - \bar{x}^{n+1})^2 \\
=& \frac{1}{n}\left[(x^{n+1} - \bar{x}^{n+1})^2 + \sum_{i=1}^n (x^i - \bar{x}^n + \bar{x}^n - \bar{x}^{n+1})^2\right] \\
=& \frac{(x^{n+1} - \bar{x}^{n+1})^2}{n} + \frac{1}{n}\sum_{i=1}^n \left[(x^i - \bar{x}^n)^2 + 2(x^i - \bar{x}^n)(\bar{x}^n - \bar{x}^{n+1}) + (\bar{x}^n - \bar{x}^{n+1})^2\right] \\
=& \frac{(x^{n+1} - \bar{x}^{n+1})^2}{n} + (\bar{x}^n - \bar{x}^{n+1})^2 + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2.
\end{aligned}$$

Substituting $\bar{x}^{n+1} = \frac{x^{n+1} + n\bar{x}^n}{n+1}$, we get

$$\begin{aligned}
\hat{\sigma}^2(x^{n+1}) =& \frac{1}{n}\left(x^{n+1} - \frac{x^{n+1} + n\bar{x}^n}{n+1}\right)^2 + \left(\bar{x}^n - \frac{n\bar{x}^n + x^{n+1}}{n+1}\right)^2 + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2 \\
=& \frac{n}{(n+1)^2}(x^{n+1} - \bar{x}^n)^2 + \frac{1}{(n+1)^2}(\bar{x}^n - x^{n+1})^2 + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2 \\
=& \frac{(x^{n+1} - \bar{x}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2.
\end{aligned}$$

■

Lemma S2 *Let $X^1, \ldots, X^n \in \mathbb{R}$ be real-valued random variables with sample mean $\bar{X}^n$. Let X^{n+1} and $\tilde{X}^{n+1}$ be two additional random variables such that*

$$|X^{n+1} - \bar{X}^n| \leq |\tilde{X}^{n+1} - \bar{X}^n| \quad a.s.$$

Then,

$$\hat{\sigma}(X^{n+1}) \leq \hat{\sigma}(\tilde{X}^{n+1}) \quad a.s.,$$

where $\hat{\sigma}(X^{n+1})$ is the sample standard deviation of $\{X^1, \ldots, X^n, X^{n+1}\}$, as in Lemma S1.

Proof This is a direct application of Lemma S1:

$$\begin{aligned}\hat{\sigma}^2(X^{n+1}) &= \frac{(X^{n+1} - \bar{X}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^{n}(X^i - \bar{X}^n)^2 \\ &\leq \frac{(\tilde{X}^{n+1} - \bar{X}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^{n}(X^i - \bar{X}^n)^2 \quad \text{[Assumption]} \\ &= \hat{\sigma}^2(\tilde{X}^{n+1}),\end{aligned}$$

and the claim follows by taking square roots. ■

Lemma S3 *Let $X^1, \ldots, X^{n+1} \in \mathbb{R}$ be real-valued random variables with sample mean $\bar{X}^{n+1}$ and sample standard deviation $\bar{S}^{n+1} > 0$. Define $Z^i = (X^i - \bar{X}^{n+1})/\bar{S}^{n+1}$ for $i \in [n+1]$. Then,*

$$|Z^i| \leq \frac{n}{\sqrt{n+1}} \quad \textit{almost surely}, \quad \forall i \in [n+1].$$

This inequality is strict if $X^1, \ldots, X^{n+1}$ are almost surely distinct.

Proof Fix any $i \in [n+1]$. Let $\bar{X}^{-i} = \frac{1}{n}\sum_{k \neq i} X^k$ be the sample mean of the random variables excluding the i-th one, and

$$\bar{S}^{-i} = \sqrt{\frac{1}{n}\sum_{k \neq i}(X^k - \bar{X}^{-i})^2}.$$

Then, it follows from Lemma S1 that

$$|Z^i| = \frac{|X^i - \bar{X}^{n+1}|}{\bar{S}^{n+1}} = \frac{\left|X^i - \frac{1}{n+1}\sum_{k=1}^{n+1} X^k\right|}{\sqrt{\frac{1}{n}\sum_{k=1}^{n+1}(X^k - \bar{X}^{n+1})^2}} = \frac{\frac{n}{n+1}|X^i - \bar{X}^{-i}|}{\sqrt{\frac{1}{n+1}(X^i - \bar{X}^{-i})^2 + (\bar{S}^{-i})^2}}.$$

But $\bar{S}^{-i} \geq 0$, so we must have

$$|Z^i| \leq \frac{\frac{n}{n+1}|X^i - \bar{X}^{-i}|}{\sqrt{\frac{1}{n+1}(X^i - \bar{X}^{-i})^2}} = \frac{n}{\sqrt{n+1}}.$$

If $X^1, \ldots, X^{n+1}$ are almost surely distinct, we know $\bar{S}^{-i} > 0$ almost surely for any $i \in [n+1]$, so this inequality becomes strict. ■

B.2 Proofs of Main Results

Proof [of Theorem 1] This is a well-known result, whose proof is included for completeness. By definition of $\hat{C}(\mathbf{X}^{n+1})$ in (5),

$$\mathbf{Y}^{n+1} \notin \hat{C}(\mathbf{X}^{n+1}) \iff S^{n+1} > \hat{Q}_{1-\alpha}(S^1, \ldots, S^n).$$

Moreover, $S^1, \ldots, S^{n+1}$ are exchangeable because $\mathbf{E}^1, \ldots, \mathbf{E}^{n+1}$ are exchangeable and Φ is fixed. Therefore, the proof is complete by applying the fundamental "quantile inflation lemma" of conformal prediction (e.g., see Angelopoulos et al. (2024)), which states

$$\mathbb{P}\left[S^{n+1} > \hat{Q}_{1-\alpha}(S^1, \ldots, S^n)\right] \leq \alpha.$$

Moreover, it is also a standard result that, if $S^1, \ldots, S^{n+1}$ are almost surely distinct,

$$\mathbb{P}\left[S^{n+1} > \hat{Q}_{1-\alpha}(S^1, \ldots, S^n)\right] \geq \alpha - \frac{1}{n+1}.$$

■

Proof [of Lemma 2] From Assumption 1, we know that $E_j^1, \ldots, E_j^{n+1}$ are almost surely distinct for every $j \in [d]$ and $0 < \hat{\sigma}_j^{\text{oracle}} < \infty$ almost surely for all $j \in [d]$, which also ensures S_{oracle}^{n+1} is well-defined. To prove this lemma, fix any $c \in \mathbb{R}$ and note that

$$\begin{aligned}
& S_{\text{oracle}}^{n+1} \leq c \\
\iff & E_j^{n+1} \leq c\, \hat{\sigma}_j^{\text{oracle}} + \hat{\mu}_j^{\text{oracle}}, \quad \forall j \in [d] \\
\iff & E_j^{n+1} - c\sqrt{\frac{1}{n}\sum_{i=1}^{n+1}\left(E_j^i - \frac{1}{n+1}\sum_{i=1}^{n+1} E_j^i\right)^2} - \frac{1}{n+1}\sum_{i=1}^{n+1} E_j^i \leq 0, \quad \forall j \in [d] \\
\iff & \frac{n}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right) - c\sqrt{\frac{1}{n}\sum_{i=1}^{n+1}\left(E_j^i - \frac{1}{n+1}\sum_{i=1}^{n+1} E_j^i\right)^2} \leq 0, \quad \forall j \in [d] \\
\iff & \underbrace{\frac{n}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right) - c\sqrt{\frac{1}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 + (\hat{\sigma}_j^{\text{obs}})^2}}_{=:g_j(\mathbf{E}^{n+1})} \leq 0, \quad \forall j \in [d]
\end{aligned}$$

where the last implication follows from Lemma S1. Everything except $\mathbf{E}^{n+1}$ here is observed, so we can solve this inequality for $\mathbf{E}^{n+1}$ coordinate-wise.

Fix any $j \in [d]$. It follows from the above that

$$g_j(\mathbf{E}^{n+1}) \leq 0 \iff \frac{n}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right) \leq c\sqrt{\frac{1}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 + (\hat{\sigma}_j^{\text{obs}})^2}. \tag{S1}$$

The solution of this inequality in E_j^{n+1} depends on the range of c. There are two cases:

1. if $c \geq 0$, then (S1) holds if and only if *either* $0 \leq E_j^{n+1} < \hat{\mu}_j^{\text{obs}}$ *or* $E_j^{n+1} \geq \hat{\mu}_j^{\text{obs}}$ and

$$\left(\frac{n}{n+1}\right)^2\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 \leq c^2\left(\frac{1}{n+1}(E_j^{n+1} - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2\right).$$

2. if $c < 0$, then (S1) holds if and only if $E_j^{n+1} < \hat{\mu}_j^{\text{obs}}$ *and*

$$\left(\frac{n}{n+1}\right)^2\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 \geq c^2\left(\tfrac{1}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 + (\hat{\sigma}_j^{\text{obs}})^2\right).$$

The solution in both cases relies on solving a quadratic equality in the form $h(x) = 0$, where:

$$h(x) := \left(\frac{n}{n+1}\right)^2 \left(x - \hat{\mu}_j^{\text{obs}}\right)^2 - c^2 \tfrac{1}{n+1} \left(x - \hat{\mu}_j^{\text{obs}}\right)^2 - c^2 (\hat{\sigma}_j^{\text{obs}})^2 = 0.$$

This equation can be written in the standard quadratic form

$$h(x) = Ax^2 + Bx + K = 0,$$

where

$$\begin{aligned}
A &= \left(\frac{n}{n+1}\right)^2 - \frac{c^2}{n+1}, \\
B &= 2\hat{\mu}_j^{\text{obs}} \left(\frac{n}{n+1}\right)^2 - 2\hat{\mu}_j^{\text{obs}} \frac{c^2}{n+1}, \\
K &= \left(\frac{n}{n+1}\right)^2 (\hat{\mu}_j^{\text{obs}})^2 - \frac{c^2}{n+1} (\hat{\mu}_j^{\text{obs}})^2 - c^2 (\hat{\sigma}_j^{\text{obs}})^2.
\end{aligned}$$

The solution set is:

$$h(x) = 0 \iff x = \begin{cases} \hat{\mu}_j^{\text{obs}} \pm \hat{\sigma}_j^{\text{obs}} |c| \sqrt{\frac{(n+1)^2}{n^2 - (n+1)c^2}}, & c^2 < \frac{n^2}{n+1}, \\ \hat{\mu}_j^{\text{obs}}, & c^2 = \frac{n^2}{n+1} \text{ and } \hat{\sigma}_j^{\text{obs}} = 0, \\ \text{no real solution}, & \text{otherwise}. \end{cases}$$

We first note that the event $\hat{\sigma}_j^{\text{obs}} = 0$ occurs if and only if $E_j^1 = \ldots = E_j^n$, which has probability zero by Assumption 1. For the rest, we note that $c^2 < \frac{n^2}{n+1}$ implies $A > 0$, so $h(x) = Ax^2 + Bx + K$ is convex for all $x \geq 0$. Analyzing the sign of $h(x)$ in the two cases shows that (S1) holds almost surely if and only if

$$\begin{cases} \hat{\mu}_j^{\text{obs}} \leq E_j^{n+1} \leq \hat{\mu}_j^{\text{obs}} + \hat{\sigma}_j^{\text{obs}} |c| \sqrt{\frac{(n+1)^2}{n^2 - (n+1)c^2}}, & c \geq 0,\ c^2 < \frac{n^2}{n+1}, \\ E_j^{n+1} < \infty, & c \geq 0,\ c^2 \geq \frac{n^2}{n+1}, \\ 0 \leq E_j^{n+1} < \hat{\mu}_j^{\text{obs}}, & c \geq 0, \\ 0 \leq E_j^{n+1} \leq \hat{\mu}_j^{\text{obs}} - \hat{\sigma}_j^{\text{obs}} |c| \sqrt{\frac{(n+1)^2}{n^2 - (n+1)c^2}}, & c < 0,\ c^2 < \frac{n^2}{n+1}, \\ \text{no solution}, & c < 0,\ c^2 \geq \frac{n^2}{n+1}. \end{cases}$$

$$\iff \begin{cases} 0 \leq E_j^{n+1} \leq \hat{\mu}_j^{\text{obs}} + \hat{\sigma}_j^{\text{obs}} |c| \sqrt{\frac{(n+1)^2}{n^2 - (n+1)c^2}}, & 0 \leq c < \frac{n}{\sqrt{n+1}}, \\ 0 \leq E_j^{n+1} < \infty, & c \geq \frac{n}{\sqrt{n+1}}, \\ 0 \leq E_j^{n+1} \leq \max\left\{0, \hat{\mu}_j^{\text{obs}} - \hat{\sigma}_j^{\text{obs}} |c| \sqrt{\frac{(n+1)^2}{n^2 - (n+1)c^2}}\right\}, & -\frac{n}{\sqrt{n+1}} < c < 0, \\ \text{no solution}, & c \leq -\frac{n}{\sqrt{n+1}}. \end{cases}$$

Interpreting the "no-solution" case as $E_j^{n+1} = 0$, we obtain precisely the expression for $\omega_j(c)$ given in the statement of the lemma. This notation adjustment is harmless because both $E_j^{n+1} = 0$ and $S_{\text{oracle}}^{n+1} \leq -n/\sqrt{n+1}$ are probability-zero events by our Assumption 1 and Lemma S3, respectively. The monotonicity of $\omega_j(c)$ follows immediately from the construction. ■

Proof [of Lemma 3] From Assumption 1, $E_j^1, \dots, E_j^{n+1}$ are almost surely distinct for every $j \in [d]$ and $0 < \hat{\sigma}_j^{\text{oracle}} < \infty$ almost surely for all $j \in [d]$. This ensures S_{oracle}^i is well-defined for all $i \in [n+1]$. Moreover, we know from Lemma S3 that

$$\left| \frac{E_j^i - \hat{\mu}_j^{\text{oracle}}}{\hat{\sigma}_j^{\text{oracle}}} \right| < \frac{n}{\sqrt{n+1}},$$

almost surely for all $i \in [n+1]$. Aggregating this inequality for all $j \in [d]$, we obtain

$$|S_{\text{oracle}}^i| = \left| \max_{1 \leq j \leq d} \frac{E_j^i - \hat{\mu}_j^{\text{oracle}}}{\hat{\sigma}_j^{\text{oracle}}} \right| \leq \max_{1 \leq j \leq d} \left| \frac{E_j^i - \hat{\mu}_j^{\text{oracle}}}{\hat{\sigma}_j^{\text{oracle}}} \right| < \max_{1 \leq j \leq d} \frac{n}{\sqrt{n+1}} = \frac{n}{\sqrt{n+1}},$$

almost surely for all $i \in [n+1]$. Recall that the quantile $\hat{Q}_{1-\alpha}^{\text{oracle}}$ is defined as

$$\hat{Q}_{1-\alpha}^{\text{oracle}} := \lceil (1-\alpha)(n+1) \rceil \text{th smallest value in } \{S_{\text{oracle}}^1, \dots, S_{\text{oracle}}^n, +\infty\},$$

so $\hat{Q}_{1-\alpha}^{\text{oracle}} < n/\sqrt{n+1}$ almost surely if $n \geq 1/\alpha - 1$ and $\hat{Q}_{1-\alpha}^{\text{oracle}} = +\infty$ otherwise. ■

Proof [of Lemma 4] Fix any $j \in [d]$ and let

$$g(t_j, z_j) := \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)}.$$

Finding the supremum of this g over $z_j \geq 0$ is a simple univariate optimization problem. Expanding g in full, we get

$$g(t_j, z_j) = \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} = \frac{t_j - \frac{1}{n+1}(z_j + n\hat{\mu}_j^{\text{obs}})}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}},$$

where the last equality follows from Lemma S1. Differentiating g with respect to z_j yields

$$g_{z_j}(t_j, z_j) = \frac{\begin{array}{c} -\frac{1}{n+1}\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2} \\ -\left(t_j - \frac{1}{n+1}(z_j + n\hat{\mu}_j^{\text{obs}})\right)\frac{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}} \end{array}}{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}.$$

Solving $g_{z_j}(t_j, z_j^*) = 0$ with respect to z_j^* gives:

$$(\hat{\sigma}_j^{\text{obs}})^2 + (t_j - \hat{\mu}_j^{\text{obs}})(z_j^* - \hat{\mu}_j^{\text{obs}}) = 0 \quad \Longrightarrow \quad z_j^* = \hat{\mu}_j^{\text{obs}} - \frac{(\hat{\sigma}_j^{\text{obs}})^2}{t_j - \hat{\mu}_j^{\text{obs}}}.$$

There are three cases here for any fixed $t_j \geq 0$:

- Case 1: if $g(t_j, z_j^*)$ is indeed the global maximum and $z_j^* > 0$, then we can just take

$$\sup_{z_j \geq 0} g(t_j, z_j) = g(t_j, z_j^*).$$

- Case 2: if $g(t_j, z_j^*)$ is not the global maximum or $z_j^* < 0$, then the supremum must occur at either $z_j = 0$ or $z_j \to \infty$ due to the fact that g_{z_j} has only one critical point. Hence,

$$\sup_{z_j \geq 0} g(t_j, z_j) = \max\left\{g(t_j, 0), \lim_{z_j \to \infty} g(t_j, z_j)\right\} = \max\left\{g(t_j, 0), -\frac{1}{\sqrt{n+1}}\right\}.$$

- Case 3: if $t_j = \hat{\mu}_j^{\text{obs}}$, then the same derivation shows that the numerator of g_{z_j} reduces to $-\frac{1}{n+1}(\hat{\sigma}_j^{\text{obs}})^2 < 0$. This implies that g is monotonically decreasing and thus we take

$$\sup_{z_j \geq 0} g(\hat{\mu}_j^{\text{obs}}, z_j) = g(\hat{\mu}_j^{\text{obs}}, 0).$$

Collecting all candidates, we obtain the closed form:

$$\sup_{z_j \geq 0} g(t_j, z_j) = \max\left\{g\left(t_j, 0\right), g\left(t_j, z_j^* \mathbb{I}\left\{z_j^* \geq 0\right\}\right), -\frac{1}{\sqrt{n+1}}\right\}.$$

■

Proof [of Proposition 6] Throughout this proof we write $\tilde{C}(\mathbf{X}^{n+1}) := \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$ for the estimated oracle prediction set defined in (11). Let $\mathcal{P}$ be any (random) partition of $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$, so that $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) = \bigsqcup_{\mathcal{R} \in \mathcal{P}} \mathcal{R}$ almost surely. Because $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \supseteq \tilde{C}(\mathbf{X}^{n+1})$ almost surely, we know

$$\begin{aligned}
\mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}) &\iff \mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}), \mathbf{Y}^{n+1} \in \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \\
&\iff \exists \mathcal{R} \in \mathcal{P} : \mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}), \mathbf{Y}^{n+1} \in \mathcal{R} \\
&\iff \exists \mathcal{R} \in \mathcal{P} : \mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}) \cap \mathcal{R}.
\end{aligned}$$

If $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ satisfies the containment on the right-hand side of (19),

$$\mathcal{R} \supseteq \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \supseteq \tilde{C}(\mathbf{X}^{n+1}) \cap \mathcal{R},$$

then

$$\begin{aligned}
\mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}) &\implies \exists \mathcal{R} \in \mathcal{P} : \mathbf{Y}^{n+1} \in \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \\
&\iff \mathbf{Y}^{n+1} \in \bigsqcup_{\mathcal{R} \in \mathcal{P}} \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \\
&\implies \mathbf{Y}^{n+1} \in \text{Rect}_{\mathcal{P}}\left(\bigsqcup_{\mathcal{R} \in \mathcal{P}} \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})\right) = \hat{C}_{\mathcal{P}}^{\text{lwc}}(\mathbf{X}^{n+1}),
\end{aligned}$$

where the last implication holds because $\mathrm{Rect}_{\mathcal{P}}(A) \supseteq A$ by definition. Since we already know that $\mathbb{P}[\mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1})] \geq 1-\alpha$, by Theorem 1 and the dual relation (12), this proves that

$$\mathbb{P}\left[\mathbf{Y}^{n+1} \in \hat{C}^{\mathrm{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})\right] \geq 1-\alpha.$$

Finally, the containment $\hat{C}^{\mathrm{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1}) \subseteq \hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$ almost surely is immediate from the definition of $\mathrm{Rect}_{\mathcal{P}}$ in (20), because

$$\hat{C}^{\mathrm{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \subseteq \mathcal{R} \subseteq \hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1}),$$

for every $\mathcal{R} \in \mathcal{P}$. ■

Proof [of Lemma 7] Fix any $\mathbf{h} \in [n+1]^d$. Suppose $\mathcal{Y}^{\mathbf{h}} \neq \emptyset$. Then $\mathcal{E}^{\mathbf{h}} \neq \emptyset$ and, by (21),

$$\mathcal{E}^{\mathbf{h}} = \bigtimes_{j=1}^{d} \mathcal{E}^{\mathbf{h}}_j, \quad \text{where} \quad \mathcal{E}^{\mathbf{h}}_j := \left[E_j^{(h_j-1)}, U_j^{h_j}\right) \neq \emptyset, \quad \forall j \in [d].$$

Fix any $\mathbf{t} \in \mathbb{R}^d_+$. Similar to the proof of Lemma 4, we have to solve a univariate optimization problem. We first notice that for any $j \in [d]$,

$$\sup_{\mathbf{z} \in \mathcal{E}^{\mathbf{h}}} \frac{t_j}{\hat{\sigma}_j(z_j)} = \sup_{z_j \in \mathcal{E}^{\mathbf{h}}_j} \frac{t_j}{\hat{\sigma}_j(z_j)} = \frac{t_j}{\inf_{z_j \in \mathcal{E}^{\mathbf{h}}_j} \hat{\sigma}_j(z_j)}.$$

By Lemma S1, we know that

$$\hat{\sigma}_j(z_j) = \sqrt{\frac{1}{n+1}(z_j - \hat{\mu}^{\mathrm{obs}}_j)^2 + (\hat{\sigma}^{\mathrm{obs}}_j)^2},$$

which is continuous in z_j, so the infimum above is attained on the closure $\overline{\mathcal{E}^{\mathbf{h}}_j} = \left[E_j^{(h_j-1)}, U_j^{h_j}\right]$; clearly $\hat{\sigma}_j(z_j)$ is minimized there at

$$z_j^* = \operatorname*{argmin}_{z_j \in \overline{\mathcal{E}^{\mathbf{h}}_j}} |z_j - \hat{\mu}^{\mathrm{obs}}_j| = \begin{cases} \hat{\mu}^{\mathrm{obs}}_j, & \text{if } \hat{\mu}^{\mathrm{obs}}_j \in \overline{\mathcal{E}^{\mathbf{h}}_j}, \\ E_j^{(h_j-1)}, & \text{if } \hat{\mu}^{\mathrm{obs}}_j < E_j^{(h_j-1)}, \\ U_j^{h_j}, & \text{if } \hat{\mu}^{\mathrm{obs}}_j > U_j^{h_j}. \end{cases}$$

Hence, $r_j(h_j) = \hat{\sigma}_j(z_j^*)$ is exactly the infimum proposed. For the second part, denote

$$g(z_j) := \frac{\hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} = \frac{\frac{1}{n+1}(z_j + n\hat{\mu}^{\mathrm{obs}}_j)}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}^{\mathrm{obs}}_j)^2 + (\hat{\sigma}^{\mathrm{obs}}_j)^2}} = \frac{\frac{1}{n+1}(z_j - \hat{\mu}^{\mathrm{obs}}_j) + \hat{\mu}^{\mathrm{obs}}_j}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}^{\mathrm{obs}}_j)^2 + (\hat{\sigma}^{\mathrm{obs}}_j)^2}},$$

and consider

$$\inf_{\mathbf{z} \in \mathcal{E}^{\mathrm{gwc}}} g(z_j) = \inf_{z_j \in [0, \omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})]} g(z_j).$$

Differentiating $g(z_j)$ in z_j, we get

$$g'(z_j) = \frac{\frac{1}{n+1}(\hat{\sigma}_j^{\text{obs}})^2 - \frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})\hat{\mu}_j^{\text{obs}}}{(\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2)\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}}.$$

Setting this to zero gives the critical point

$$z_j^* = \hat{\mu}_j^{\text{obs}} + \frac{(\hat{\sigma}_j^{\text{obs}})^2}{\hat{\mu}_j^{\text{obs}}} > 0.$$

For $\epsilon > 0$, we have

$$\begin{aligned}\operatorname{sign}(g'(z_j^* + \epsilon)) &= \operatorname{sign}\left(-\tfrac{1}{n+1}\epsilon\hat{\mu}_j^{\text{obs}}\right) = -1,\\ \operatorname{sign}(g'(z_j^* - \epsilon)) &= \operatorname{sign}\left(\tfrac{1}{n+1}\epsilon\hat{\mu}_j^{\text{obs}}\right) = +1.\end{aligned}$$

This implies that $g(z_j^*)$ is always a local maximum. Hence, if $z_j^* \in [0, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})]$, then

$$\inf_{z_j \in [0, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})]} g(z_j) = \min\left\{g(0), \lim_{z_j \uparrow \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})} g(z_j)\right\};$$

if $z_j^* > \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})$, then g is nondecreasing on $[0, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})]$ and

$$\inf_{z_j \in [0, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})]} g(z_j) = g(0).$$

But in either case, we only need to compare the boundaries in order to find the infimum. Therefore, it follows that

$$\inf_{z_j \in [0, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})]} g(z_j) = \min\left\{g(0), \lim_{z_j \uparrow \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})} g(z_j)\right\}.$$

This is true for every $j \in [d]$, so we must have

$$\begin{aligned}\hat{\Phi}_{\mathbf{h}}^{\text{lwc}}(\mathbf{t}) &= \max_{1 \le j \le d}\left\{\sup_{\mathbf{z} \in \mathcal{E}^{\mathbf{h}}} \frac{t_j}{\hat{\sigma}_j(z_j)} - \inf_{\mathbf{z} \in \mathcal{E}^{\text{gwc}}} g(z_j)\right\}\\ &= \max_{1 \le j \le d}\left\{\frac{t_j}{r_j(h_j)} - \min\left\{\frac{\hat{\mu}_j(0)}{\hat{\sigma}_j(0)}, \lim_{z_j \uparrow \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})} \frac{\hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)}\right\}\right\}.\end{aligned}$$

■

Proof [of Lemma 8] Fix any $j \in [d]$ and suppose $\mathcal{Y}^{\mathbf{h}^*} \neq \emptyset$. There are two parts to this lemma, which we prove separately. Throughout, for any $k \in [n+1]$ we let $\mathbf{h}^*(j,k)$ denote the unique multi-index in $\text{Row}_j(\mathbf{h}^*)$ whose j-th coordinate equals k; that is, $h_j^*(j,k) = k$ and $h_l^*(j,k) = h_l^*$ for all $l \neq j$.

Part I. We aim to prove

$$\sup_{\mathbf{h}\in[n+1]^d} B_j^{\mathbf{h}} = \sup_{\mathbf{h}\in\mathrm{Row}_j(\mathbf{h}^*)} B_j^{\mathbf{h}} = B_j^{\mathbf{h}^{[j]}}. \tag{S2}$$

We first notice that the first supremum can be decomposed into suprema over surfaces:

$$\sup_{\mathbf{h}\in[n+1]^d} B_j^{\mathbf{h}} = \sup_{k\in[n+1]} \sup_{\mathbf{h}\in\mathrm{Surface}_j(k)} B_j^{\mathbf{h}},$$

where

$$\mathrm{Surface}_j(k) := \left\{\mathbf{h}\in[n+1]^d : h_j = k\right\}.$$

Fix any $k\in[n+1]$ and note that $\mathbf{h}^*(j,k)\in\mathrm{Surface}_j(k)\cap\mathrm{Row}_j(\mathbf{h}^*)$. Since this holds for every $k\in[n+1]$, to prove (S2) the main challenge is to show that:

$$\sup_{\mathbf{h}\in\mathrm{Surface}_j(k)} B_j^{\mathbf{h}} = B_j^{\mathbf{h}^*(j,k)}. \tag{S3}$$

To prove (S3), recall from the definition in (22) and Lemma 7 that

$$\hat{\Phi}_{\mathbf{h}}^{\mathrm{lwc}}(\mathbf{t}) = \max_{1\le l\le d}\left\{\frac{t_l}{r_l(h_l)} - c_l\right\},$$

where

$$r_l(h_l) = \min\left\{\hat{\sigma}_l(z_l) : z_l\in\overline{\mathcal{E}_l^{\mathbf{h}}}\right\}, \quad \overline{\mathcal{E}_l^{\mathbf{h}}} := \left[E_l^{(h_l-1)}, U_l^{h_l}\right], \qquad c_l = \inf_{\mathbf{z}\in\mathcal{E}^{\mathrm{gwc}}} \frac{\hat{\mu}_l(z_l)}{\hat{\sigma}_l(z_l)}.$$

Without loss of generality, assume $\mathcal{Y}^{\mathbf{h}}\neq\emptyset$ for all $\mathbf{h}\in\mathrm{Surface}_j(k)$. This does not affect the final result because $\mathcal{Y}^{\mathbf{h}}=\emptyset$ leads to $B_j^{\mathbf{h}}=0$, which can be safely excluded from the supremum calculation. Then notice that, for all $\mathbf{h}\in\mathrm{Surface}_j(k)$, we have $r_j(h_j)=r_j(k)$ and $r_l(h_l)\ge\hat{\sigma}_l^{\mathrm{obs}} = r_l(h_l^*)$ for all $l\neq j$, the latter because $\hat{\mu}_l^{\mathrm{obs}}\in\overline{\mathcal{E}_l^{\mathbf{h}^*}}$ by the definition of the mean index $\mathbf{h}^*$. Therefore, for any fixed $\mathbf{t}$,

$$\hat{\Phi}_{\mathbf{h}}^{\mathrm{lwc}}(\mathbf{t}) = \max_{1\le l\le d}\left\{\frac{t_l}{r_l(h_l)} - c_l\right\} \le \max\left(\max_{l\neq j}\left\{\frac{t_l}{\hat{\sigma}_l^{\mathrm{obs}}} - c_l\right\}, \frac{t_j}{r_j(k)} - c_j\right) = \hat{\Phi}_{\mathbf{h}^*(j,k)}^{\mathrm{lwc}}(\mathbf{t}).$$

Because this holds pointwise for all $\mathbf{t}$, we can extend the inequality to the corresponding conformal quantile and conclude that, almost surely for all $\mathbf{h}\in\mathrm{Surface}_j(k)$,

$$\hat{Q}_{1-\alpha}^{\mathrm{lwc}}(\mathcal{Y}^{\mathbf{h}}) \le \hat{Q}_{1-\alpha}^{\mathrm{lwc}}(\mathcal{Y}^{\mathbf{h}^*(j,k)})$$

and therefore, by the monotonicity of the function ω_j from Lemma 2,

$$\omega_j(\hat{Q}_{1-\alpha}^{\mathrm{lwc}}(\mathcal{Y}^{\mathbf{h}})) \le \omega_j(\hat{Q}_{1-\alpha}^{\mathrm{lwc}}(\mathcal{Y}^{\mathbf{h}^*(j,k)})).$$

Therefore, it follows that, for all $\mathbf{h}\in\mathrm{Surface}_j(k)$,

$$\begin{aligned}
B_j^{\mathbf{h}} &= \begin{cases} \min\left\{U_j^k, \omega_j(\hat{Q}_{1-\alpha}^{\mathrm{lwc}}(\mathcal{Y}^{\mathbf{h}}))\right\}, & \text{if } U_j^k,\ \omega_j(\hat{Q}_{1-\alpha}^{\mathrm{lwc}}(\mathcal{Y}^{\mathbf{h}})) > E_j^{(k-1)}, \\ 0, & \text{otherwise,} \end{cases} \\
&\le \begin{cases} \min\{U_j^k, \omega_j(\hat{Q}_{1-\alpha}^{\mathrm{lwc}}(\mathcal{Y}^{\mathbf{h}^*(j,k)}))\}, & \text{if } U_j^k,\ \omega_j(\hat{Q}_{1-\alpha}^{\mathrm{lwc}}(\mathcal{Y}^{\mathbf{h}^*(j,k)})) > E_j^{(k-1)}, \\ 0, & \text{otherwise,} \end{cases} \\
&= B_j^{\mathbf{h}^*(j,k)}.
\end{aligned}$$

This completes the proof of (S3).

The second equality in (S2) is trivial. We can easily see that for any $k \in [n]$, if $B_j^{\mathbf{h}^*(j,k+1)} \neq 0$, then

$$U_j^{k+1},\ \omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,k+1)})) \geq E_j^{(k)} \geq \min\left\{E_j^{(k)}, \omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})\right\} = U_j^k.$$

Hence, it follows that $B_j^{\mathbf{h}^*(j,k+1)} \geq B_j^{\mathbf{h}^*(j,k)}$ regardless of whether $B_j^{\mathbf{h}^*(j,k)} = 0$ or not. Then finding the supremum among all $k \in [n+1]$ simplifies to finding the rightmost nonzero $B_j^{\mathbf{h}^*(j,k)}$, which is exactly how $B_j^{\mathbf{h}^{[j]}}$ is defined.

Part II. In this part, we aim to show that if $B_j^{\mathbf{h}^*(j,m_j)} = 0$ for some $m_j \geq h_j^*$, then

$$B_j^{\mathbf{h}^*(j,N)} = 0, \quad \forall N \geq m_j.$$

There are two cases that could lead to $B_j^{\mathbf{h}^*(j,m_j)} = 0$:

- If $U_j^{m_j} = \min\left\{E_j^{(m_j)}, \omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})\right\} \leq E_j^{(m_j-1)}$, then $\omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha}) \leq E_j^{(m_j-1)} \leq E_j^{(N-1)}$ for any $N \geq m_j$ by the nature of order statistics. Hence, $U_j^N \leq E_j^{(N-1)}$ for all $N \geq m_j$, which further implies $B_j^{\mathbf{h}^*(j,N)} = 0$ for all $N \geq m_j$.

- If $\omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,m_j)})) \leq E_j^{(m_j-1)}$, then we claim that

$$\omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,N)})) \leq \omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,m_j)})) \leq E_j^{(m_j-1)} \leq E_j^{(N-1)},$$

  which then implies $B_j^{\mathbf{h}^*(j,N)} = 0$ for all $N \geq m_j$. This is actually trivial because $m_j \geq h_j^*$ implies that both $\overline{\mathcal{E}_j^{\mathbf{h}^*(j,m_j)}}$ and $\overline{\mathcal{E}_j^{\mathbf{h}^*(j,N)}}$ lie to the right of $\hat{\mu}_j^{\mathrm{obs}}$, with the latter no closer to it than the former, so that

$$r_j(m_j) = \min_{z_j \in \overline{\mathcal{E}_j^{\mathbf{h}^*(j,m_j)}}} \hat{\sigma}_j(z_j) \leq \min_{z_j \in \overline{\mathcal{E}_j^{\mathbf{h}^*(j,N)}}} \hat{\sigma}_j(z_j) = r_j(N),$$

  by Lemma S2 and the definition of $\mathcal{E}_j^{\mathbf{h}}$. This then leads to

$$\begin{aligned}
&\hat{\Phi}^{\mathrm{lwc}}_{\mathbf{h}^*(j,m_j)}(\mathbf{t}) \geq \hat{\Phi}^{\mathrm{lwc}}_{\mathbf{h}^*(j,N)}(\mathbf{t}), \quad \forall \mathbf{t} \\
&\implies \hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,N)}) \leq \hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,m_j)}) \\
&\implies \omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,N)})) \leq \omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{\mathbf{h}^*(j,m_j)})).
\end{aligned}$$

■

Appendix C. Additional Numerical Results

We examine how coordinate heterogeneity, dependence, calibration size, noise tails, and residual transformations affect the coverage and efficiency of TSCP. Comparisons with the population oracle and the unenclosed local union separate the statistical and computational effects of its approximations. Real-data marginal coverage and search-regime measurements complement the results in Section 4.

C.1 Additional Noise-Distribution Experiments

Tables S3–S7 report coverage and normalized residual-space volume for five noise laws. Gaussian errors contrast unequal and equal coordinate scales; Laplace, mixture, and Gamma errors vary the tail shape and symmetry under heterogeneous scaling. The Laplace distributions are centered at zero, with the stated scale parameter; the Gamma parameterization specifies shape followed by scale.

Gaussian Noise (Heterogeneous)

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Gaussian)	Emp. Copula	0.714 (0.079)	1.68e+10(1.16e+10)
	Point CHR	0.933 (0.066)	4.04e+12(1.38e+13)
	TSCP (Ours)	0.910 (0.053)	**1.83e+11(2.74e+11)**
	Unscaled Max	0.908 (0.047)	2.22e+13(4.77e+13)
(50, 10, Gaussian)	Emp. Copula	0.816 (0.057)	3.64e+10(2.22e+10)
	Point CHR	0.919 (0.057)	4.85e+11(1.72e+12)
	TSCP (Ours)	0.904 (0.047)	**9.42e+10(8.82e+10)**
	Unscaled Max	0.897 (0.043)	1.16e+13(1.10e+13)
(100, 10, Gaussian)	Emp. Copula	0.819 (0.039)	2.58e+10(9.90e+09)
	Point CHR	0.901 (0.041)	9.74e+10(9.22e+10)
	TSCP (Ours)	0.903 (0.034)	**6.59e+10(3.43e+10)**
	Unscaled Max	0.908 (0.027)	1.09e+13(6.93e+12)
(300, 10, Gaussian)	Emp. Copula	0.872 (0.019)	3.73e+10(8.11e+09)
	Point CHR	0.899 (0.025)	5.60e+10(2.05e+10)
	TSCP (Ours)	0.899 (0.018)	**4.93e+10(1.17e+10)**
	Unscaled Max	0.901 (0.020)	8.68e+12(3.29e+12)
(500, 10, Gaussian)	Emp. Copula	0.886 (0.018)	4.16e+10(8.27e+09)
	Point CHR	0.900 (0.022)	5.13e+10(1.49e+10)
	TSCP (Ours)	0.901 (0.016)	**4.81e+10(9.67e+09)**
	Unscaled Max	0.903 (0.016)	8.54e+12(2.46e+12)

Table S3: Performance with $d = 10$ outcome variables, as in Figure 3. Results are averaged over 200 repetitions, with one standard deviation in parentheses. The target coverage is 0.9; estimates below this target are highlighted in red. Bold values indicate the smallest mean residual-space volume among TSCP, Point CHR, and Unscaled Max. TSCP has the smallest volume among these methods in the displayed settings.

Gaussian Noise (Homogeneous)

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Unit Gaussian)	Emp. Copula	0.714 (0.079)	4.64e+03(3.20e+03)
	Point CHR	0.933 (0.066)	1.11e+06(3.80e+06)
	TSCP (Ours)	0.910 (0.053)	5.04e+04(7.56e+04)
	Unscaled Max	0.904 (0.054)	**2.19e+04(1.94e+04)**
(50, 10, Unit Gaussian)	Emp. Copula	0.816 (0.057)	1.00e+04(6.12e+03)
	Point CHR	0.919 (0.057)	1.34e+05(4.75e+05)
	TSCP (Ours)	0.904 (0.047)	2.60e+04(2.43e+04)
	Unscaled Max	0.901 (0.045)	**1.68e+04(1.15e+04)**
(100, 10, Unit Gaussian)	Emp. Copula	0.819 (0.039)	7.11e+03(2.73e+03)
	Point CHR	0.901 (0.041)	2.68e+04(2.54e+04)
	TSCP (Ours)	0.903 (0.034)	1.82e+04(9.46e+03)
	Unscaled Max	0.900 (0.035)	**1.50e+04(7.38e+03)**
(300, 10, Unit Gaussian)	Emp. Copula	0.872 (0.019)	1.03e+04(2.23e+03)
	Point CHR	0.899 (0.025)	1.54e+04(5.66e+03)
	TSCP (Ours)	0.899 (0.018)	1.36e+04(3.23e+03)
	Unscaled Max	0.898 (0.018)	**1.28e+04(3.07e+03)**
(500, 10, Unit Gaussian)	Emp. Copula	0.886 (0.018)	1.15e+04(2.28e+03)
	Point CHR	0.900 (0.022)	1.41e+04(4.11e+03)
	TSCP (Ours)	0.901 (0.016)	1.33e+04(2.67e+03)
	Unscaled Max	0.899 (0.017)	**1.26e+04(2.50e+03)**

Table S4: Performance with $d = 10$ outcome variables and homogeneous Gaussian noise, as in Figure 4. Other details are as in Table S3. Unscaled Max has the smallest volume among methods with near-nominal coverage, while TSCP remains competitive.

Laplace Noise

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Laplace)	Emp. Copula	0.712 (0.079)	3.82e+12(4.91e+12)
	Point CHR	0.938 (0.060)	5.57e+16(2.71e+17)
	TSCP (Ours)	0.908 (0.060)	**3.14e+14(9.93e+14)**
	Unscaled Max	0.897 (0.053)	6.36e+15(1.99e+16)
(50, 10, Laplace)	Emp. Copula	0.819 (0.056)	1.36e+13(1.75e+13)
	Point CHR	0.917 (0.050)	6.64e+14(1.81e+15)
	TSCP (Ours)	0.906 (0.045)	**7.32e+13(1.60e+14)**
	Unscaled Max	0.901 (0.041)	3.34e+15(5.12e+15)
(100, 10, Laplace)	Emp. Copula	0.811 (0.039)	5.96e+12(3.94e+12)
	Point CHR	0.896 (0.039)	4.74e+13(6.07e+13)
	TSCP (Ours)	0.898 (0.030)	**2.64e+13(1.94e+13)**
	Unscaled Max	0.895 (0.029)	1.77e+15(1.78e+15)
(300, 10, Laplace)	Emp. Copula	0.871 (0.020)	1.11e+13(4.03e+12)
	Point CHR	0.895 (0.023)	2.01e+13(1.12e+13)
	TSCP (Ours)	0.896 (0.017)	**1.74e+13(6.40e+12)**
	Unscaled Max	0.896 (0.017)	1.38e+15(7.08e+14)
(500, 10, Laplace)	Emp. Copula	0.882 (0.016)	1.29e+13(4.10e+12)
	Point CHR	0.896 (0.021)	1.83e+13(8.95e+12)
	TSCP (Ours)	0.896 (0.016)	**1.64e+13(5.68e+12)**
	Unscaled Max	0.897 (0.017)	1.36e+15(5.51e+14)

Table S5: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, and noise following a *Laplace* distribution; i.e., $\epsilon_j \sim \text{Laplace}(10 - j + 1)$ for all $j \in [10]$. Other details are as in Table S3.

Mixture Noise

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Mixed)	Emp. Copula	0.715 (0.075)	2.69e+11(3.04e+11)
	Point CHR	0.935 (0.062)	2.56e+15(1.98e+16)
	TSCP (Ours)	0.913 (0.051)	**9.53e+12(3.12e+13)**
	Unscaled Max	0.903 (0.052)	3.46e+13(6.74e+13)
(50, 10, Mixed)	Emp. Copula	0.821 (0.058)	7.07e+11(5.96e+11)
	Point CHR	0.925 (0.048)	6.10e+13(3.06e+14)
	TSCP (Ours)	0.906 (0.044)	**2.57e+12(3.91e+12)**
	Unscaled Max	0.901 (0.044)	1.98e+13(1.68e+13)
(100, 10, Mixed)	Emp. Copula	0.819 (0.035)	3.82e+11(1.98e+11)
	Point CHR	0.898 (0.042)	2.76e+12(4.47e+12)
	TSCP (Ours)	0.900 (0.029)	**1.23e+12(7.54e+11)**
	Unscaled Max	0.905 (0.028)	1.65e+13(9.89e+12)
(300, 10, Mixed)	Emp. Copula	0.874 (0.022)	6.47e+11(2.25e+11)
	Point CHR	0.902 (0.025)	1.27e+12(7.23e+11)
	TSCP (Ours)	0.900 (0.019)	**9.05e+11(3.48e+11)**
	Unscaled Max	0.900 (0.020)	1.36e+13(5.22e+12)
(500, 10, Mixed)	Emp. Copula	0.886 (0.016)	7.26e+11(1.82e+11)
	Point CHR	0.897 (0.020)	1.01e+12(5.42e+11)
	TSCP (Ours)	0.898 (0.014)	**8.17e+11(2.05e+11)**
	Unscaled Max	0.902 (0.016)	1.35e+13(3.73e+12)

Table S6: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, and noise following a *Mixture* distribution; i.e., $\epsilon_j \sim \frac{1}{2}\text{Laplace}(10 - j + 1) + \frac{1}{2}\mathcal{N}(0, (10 - j + 1)^2)$ for all $j \in [10]$. Other details are as in Table S3.

Gamma Noise

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Gamma)	Emp. Copula	0.714 (0.067)	5.33e-02(6.85e-02)
	Point CHR	0.929 (0.058)	7.55e+02(4.32e+03)
	TSCP (Ours)	0.904 (0.056)	**3.77e+00(1.59e+01)**
	Unscaled Max	0.898 (0.051)	2.69e+15(6.18e+15)
(50, 10, Gamma)	Emp. Copula	0.826 (0.048)	2.20e-01(2.78e-01)
	Point CHR	0.927 (0.049)	5.01e+01(3.36e+02)
	TSCP (Ours)	0.910 (0.041)	**1.15e+00(2.56e+00)**
	Unscaled Max	0.902 (0.039)	1.94e+15(3.54e+15)
(100, 10, Gamma)	Emp. Copula	0.818 (0.038)	1.04e-01(1.01e-01)
	Point CHR	0.906 (0.038)	1.28e+00(2.38e+00)
	TSCP (Ours)	0.900 (0.030)	**4.65e-01(5.82e-01)**
	Unscaled Max	0.897 (0.030)	9.67e+14(7.86e+14)
(300, 10, Gamma)	Emp. Copula	0.873 (0.019)	1.75e-01(1.15e-01)
	Point CHR	0.895 (0.024)	3.74e-01(3.70e-01)
	TSCP (Ours)	0.898 (0.018)	**2.67e-01(1.69e-01)**
	Unscaled Max	0.895 (0.018)	7.04e+14(3.45e+14)
(500, 10, Gamma)	Emp. Copula	0.885 (0.015)	1.89e-01(1.06e-01)
	Point CHR	0.896 (0.020)	3.20e-01(2.33e-01)
	TSCP (Ours)	0.897 (0.016)	**2.36e-01(1.35e-01)**
	Unscaled Max	0.893 (0.015)	6.37e+14(2.53e+14)

Table S7: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, and noise following a *Gamma* distribution; i.e., $\epsilon_j \sim \text{Gamma}(1, 10-j+1)$ for all $j \in [10]$. Other details are as in Table S3.

C.2 Dependence Strength

We vary the equicorrelation of the Gaussian noise over $\rho \in \{0, 0.3, 0.6, 0.9\}$ while retaining marginal standard deviations $(10, 9, \ldots, 1)$ and a calibration size of $n = 100$. Figure S1 shows that TSCP's empirical joint coverage is stable across different values of ρ. Its average residual-space volume decreases from 6.37×10^{10} at $\rho = 0$ to 9.47×10^{9} at $\rho = 0.9$. Point CHR and Unscaled Max also remain close to the target. Emp. Copula improves as the correlation strengthens but remains below the nominal level throughout this test. The experiment therefore does not indicate a coverage failure of TSCP under strong equicorrelation, while also showing that our method adapts well to varying dependence structures.

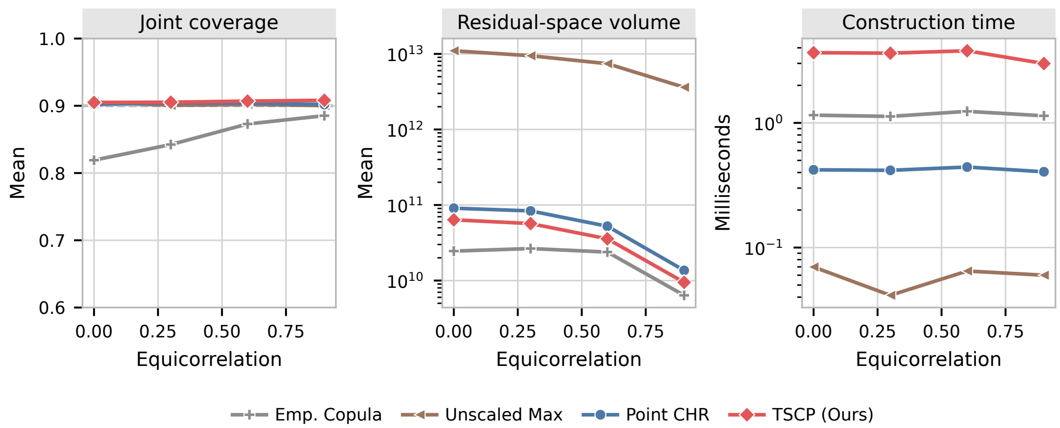


Figure S1: Sensitivity to equicorrelated Gaussian noise at $d = 10$ and $n = 100$. The three panels report joint coverage, mean residual-space volume, and mean construction time. Results average 200 repetitions; the dashed line marks the target coverage.

C.3 Severity of Coordinate Heterogeneity

To isolate heterogeneity across outcomes, we use independent Gaussian noise and choose ten coordinate scales geometrically spaced between r and 1, where $r \in \{1, 2, 5, 10, 20\}$. The calibration size is fixed at $n = 100$. Figure S2 plots joint coverage and mean residual-space volume against the scale ratio r and shows that the coverage of TSCP, Point CHR, and Unscaled Max is essentially unchanged as r grows. At $r = 1$, Unscaled Max produces slightly smaller volumes than TSCP, consistent with the unit-scale comparison in Figure 4. As r grows, Unscaled Max's volume increases more steeply than the volumes of the coordinate-adaptive methods. This widening gap is the expected benefit of coordinate adaptation: a single maximum-residual threshold becomes increasingly inefficient when one scale must be used for all coordinates.

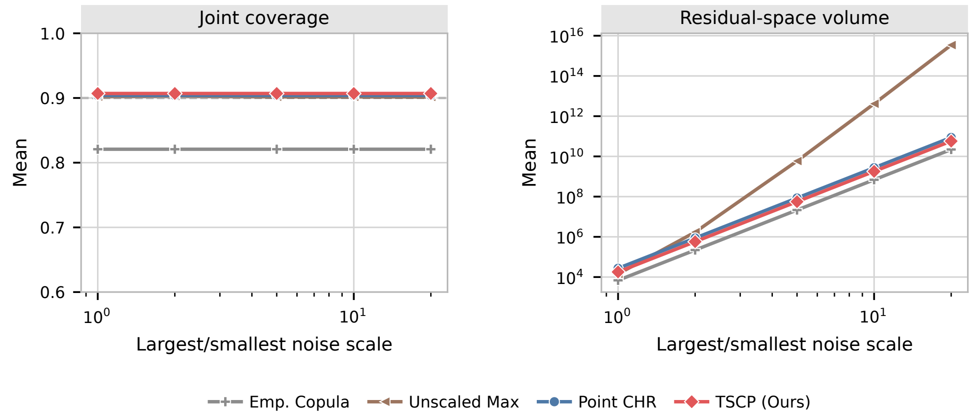


Figure S2: Sensitivity to the ratio between the largest and smallest outcome noise scales, with independent Gaussian errors and $n = 100$. The scales are geometrically spaced, and both axes of the volume panel are logarithmic. Results average 200 repetitions.

C.4 Sensitivity to the Target Coverage Level

Figure S3 repeats the dependent Gaussian experiment at nominal coverage levels 0.8, 0.9, and 0.95, with $n = 100$ and $\rho = 0.6$. TSCP achieves empirical coverages 0.811, 0.907, and 0.954. As expected, increasing the target level increases all region volumes. The relative ordering is stable: TSCP is smaller than Point CHR and dramatically smaller than Unscaled Max, while Emp. Copula is smaller but below the nominal coverage at every displayed target.

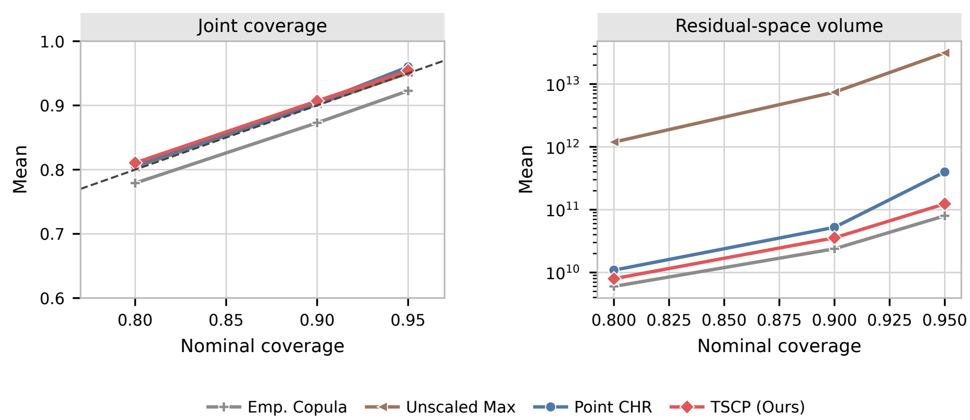
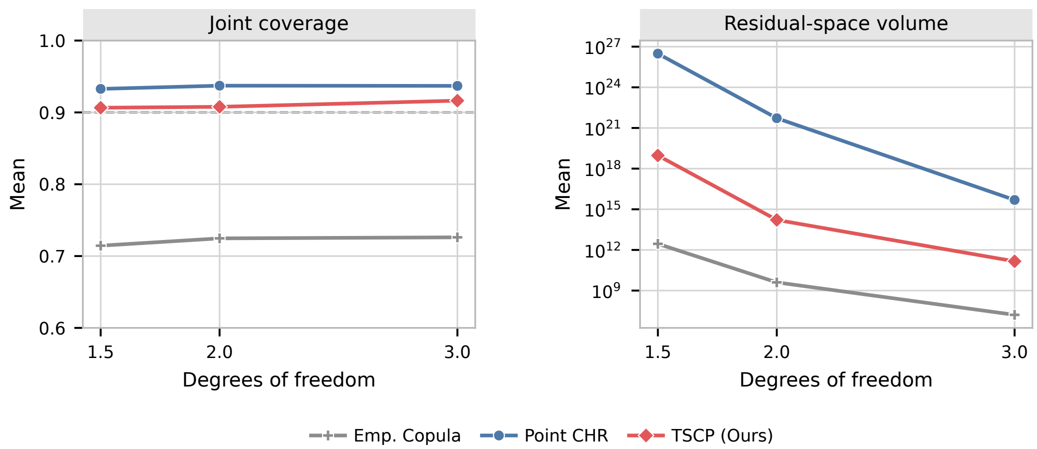


Figure S3: Sensitivity to the nominal coverage level in the $d = 10$ equicorrelated Gaussian experiment with $n = 100$ and $\rho = 0.6$. The dashed diagonal in the left panel is exact nominal calibration. Results average 200 repetitions.

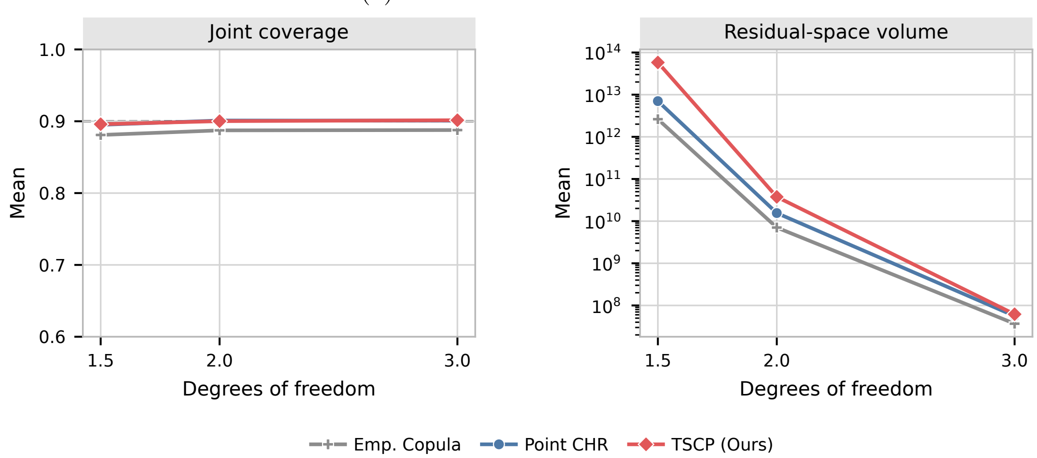
C.5 Heavy-Tailed Noise

We examine independent Student t noise with $d = 10$, a common unit scale across coordinates, and degrees of freedom $\nu \in \{1.5, 2, 3\}$. The cases $\nu = 1.5$ and $\nu = 2$ have infinite variance and therefore lie outside Assumption 1, which is invoked by the validity theorem as well as the moment-based motivation; $\nu = 3$ has finite variance but remains heavy-tailed.

Figure S4 presents the results averaged over 200 repetitions. Coverage remains near the target while interval volumes grow substantially as the degrees of freedom decrease. Point CHR has larger volume at the smaller calibration size but can outperform TSCP at the larger size.



(a) Calibration size $n = 30$.



(b) Calibration size $n = 500$.

Figure S4: Performance under Student t noise with varying degrees of freedom, using calibration sizes (a) $n = 30$ and (b) $n = 500$. The target coverage is 0.9. TSCP has near-nominal empirical coverage in these settings; the infinite-variance cases are outside the stated assumptions. Point CHR produces larger volumes at the smaller calibration size but can outperform TSCP at the larger size. Downward triangles at the coverage-axis boundary indicate values below 0.6.

C.6 Exchangeable Contamination Stress Test

To test sensitivity to atypically large residuals, we contaminate the Gaussian experiment at rates $\varepsilon \in \{0, 0.01, 0.05, 0.10\}$. Independently for each observation, all ten noise coordinates are multiplied by 10 with probability ε; the same exchangeable mechanism is used for training, calibration, and test data. The calibration size is $n = 100$. We report median rather than mean volume because the contaminated distribution produces a small number of extreme regions.

Figure S5 shows that TSCP coverage stays near nominal regardless of contamination, but its median volume grows from 5.38×10^{10} to 1.10×10^{17}. Point CHR and Unscaled Max exhibit the same qualitative inflation, and Emp. Copula undercovers at every contamination level. Thus, the conformal calibration is comparatively stable in coverage, whereas the sample mean and standard deviation used for coordinate scaling are not robust efficiency summaries. Replacing them by robust estimators would change the construction and its analysis and requires a separate validity argument.

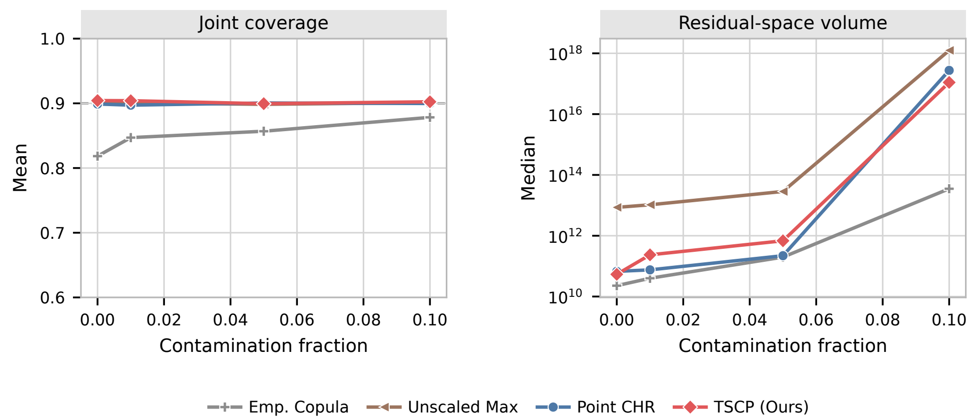


Figure S5: Row-wise contamination stress test with $d = 10$ and $n = 100$. A contaminated observation has every noise coordinate multiplied by 10. The right panel reports median residual-space volume over 200 repetitions on a logarithmic scale.

C.7 Additional CQR Sensitivity Analyses

Base-interval width. The main CQR comparison uses base-interval miscoverage 0.5. We repeat it at 0.3, 0.5, 0.7, and 0.9, fixing $n = 100$ and final miscoverage 0.1. Figure S6 shows that TSCP with capped residuals and Unscaled Max behave similarly, while Emp. Copula undercovers. CQHR remains close to nominal coverage but its volume is highly sensitive to the fitted base interval in this design.

Shifted scores. An alternative transformation replaces the signed CQR residuals E_j by $E_j + C$, apply TSCP, and subtract C from the final coordinate adjustment. We test $C \in \{100, 300, 1000\}$. As shown in Figure S7, the recorded TSCP rectangle is numerically invariant to C and, more importantly, agrees exactly with the TSCP-GWC rectangle: the maximum coordinate-length difference is zero in all 30 repetitions for every C. The proposed

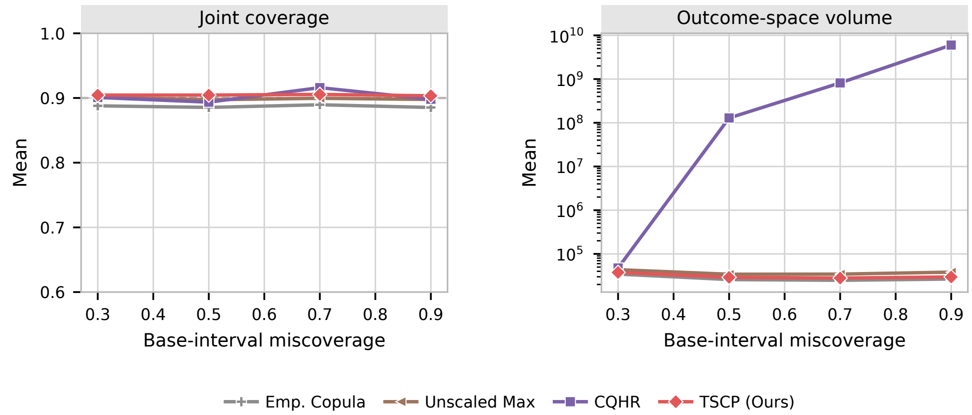


Figure S6: CQR sensitivity to the base-interval miscoverage with $d = 3$, $n = 100$, and final target coverage 0.9. TSCP uses capped scores, CQHR uses its native construction, and results average 30 independent repetitions.

explanation is that a large additive constant inflates the LWC score bound in Section 3.4, so the global-worst-case bound determines the final rectangle. Equality of the recorded rectangles is empirical evidence, not a general shift-invariance result. Capped residuals $\max\{E_j, 0\}$ avoid choosing C and are used for the main comparison, with the limitations described in Section 4.3.

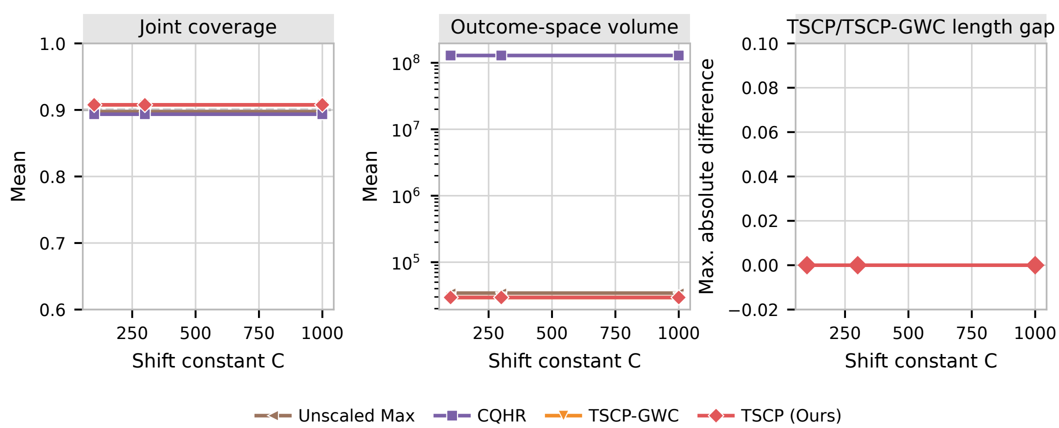


Figure S7: Sensitivity of shifted CQR scores to the additive constant C. The first two panels report coverage and mean outcome-space volume; the third reports the maximum coordinate-length difference between shifted TSCP and TSCP-GWC. Results use $d = 3$, $n = 100$, and 30 independent repetitions.

C.8 Low-Dimensional Shape-Template Baseline

We compare with the convex shape-template approach of Tumu et al. (2024). We use the public `conformal-region-designer` components: a kernel density estimate on a grid of size 20 per dimension, mean-shift clustering, and an axis-aligned hyperrectangular template. The signed calibration residuals are divided equally between fitting the density and template and conformalizing its inflation. We use a score-consistent hyperrectangle implementation and impose an aspect-ratio floor of 0.05 when a very small fitted cluster is degenerate along one coordinate. This detail keeps the conformal inflation finite.

We use $d = 2$ with independent Gaussian scales $(2, 1)$ and calibration sizes $n \in \{30, 50, 100, 200\}$. Each of the 100 repetitions uses 2,400 training observations, 600 test observations, and a separately generated calibration sample. Shape-template volume is divided by 2^d to match the residual-space volume convention used for the symmetric rectangular methods. Figure S8 shows that TSCP has smaller volume than the shape-template method at every calibration size, and its construction time is more than two orders of magnitude shorter for this implementation.

This is deliberately a low-dimensional comparison. A Cartesian KDE grid has 20^d locations and is therefore impractical at the $d = 10$ dimension used in the main absolute-residual comparisons. This limitation concerns the chosen grid implementation, not every shape-template construction.

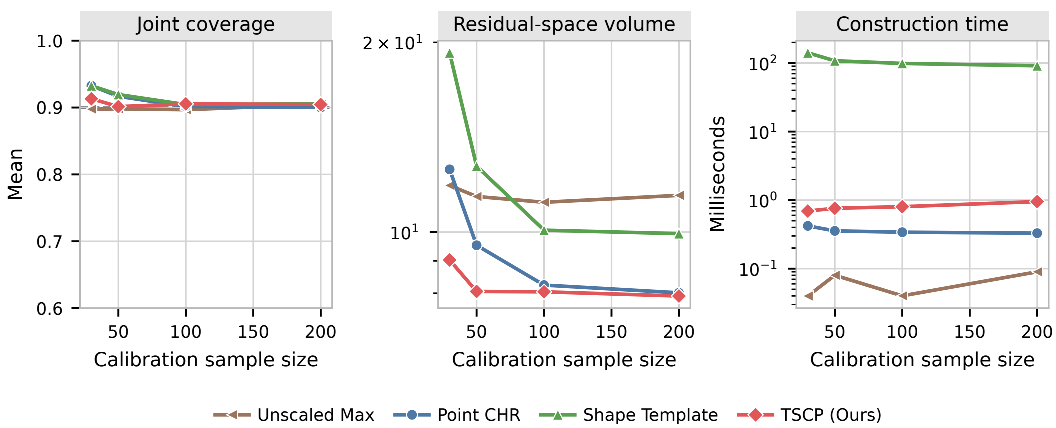


Figure S8: Comparison with a KDE/mean-shift hyperrectangular shape template for $d = 2$ Gaussian residuals with scales $(2, 1)$. Volume follows the same 2^{-d} normalization as the other residual-space experiments. Results average 100 independent repetitions. The shape-template method is slower and has larger volume than TSCP at every calibration size.

C.9 Oracle Approximation and Computational Scaling

We examine how the practical constructions approximate the population oracle under independent Laplace noise with coordinate scales $(d, d-1, \ldots, 1)$ and varying calibration size.

The additional methods considered in this section include (i) *Naive*, the heuristic plug-in rescaling method that uses location and scale parameters estimated directly from the calibration data without further adjustments, summarized in Algorithm S3; (ii) *TSCP-S*, the inefficient data-splitting variant summarized in Algorithm S4; (iii) *TSCP-GWC*, our conservative global-worst-case method summarized in Algorithm S5; (iv) *TSCP-LWC*, the aggregated union described in Section 3.4 but not enclosed by a rectangle.

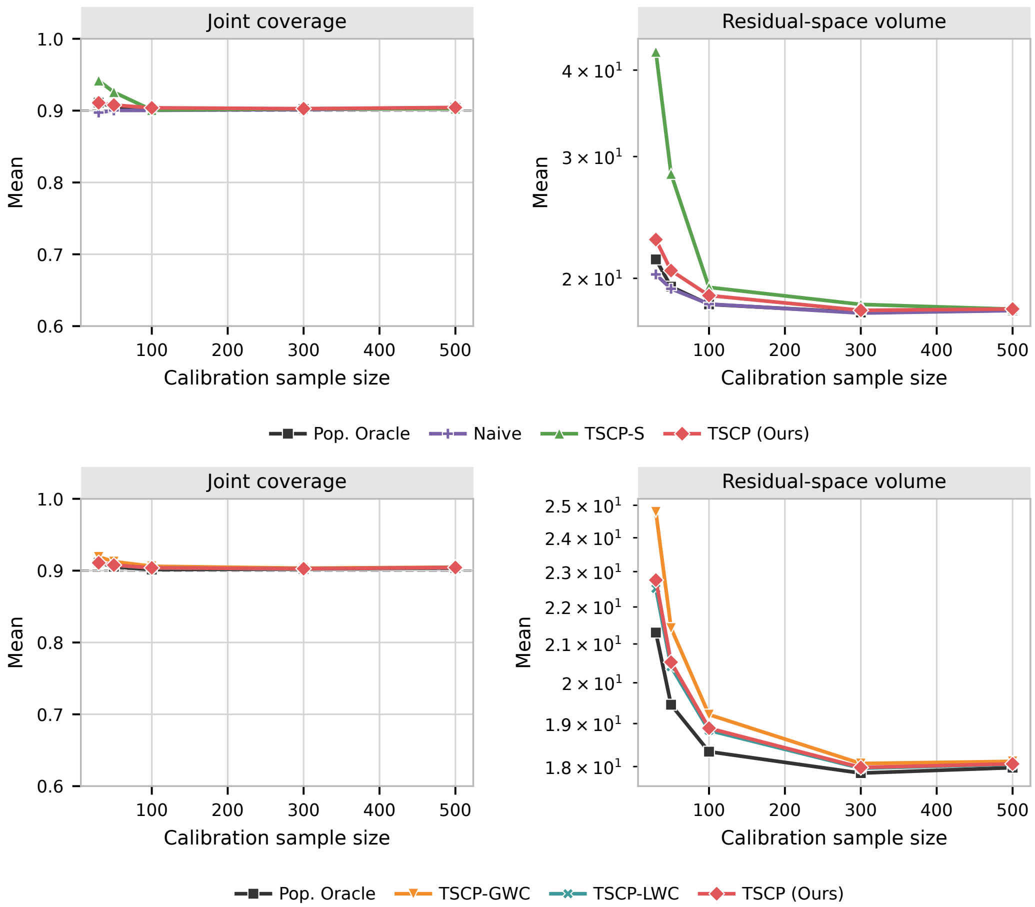


Figure S9: Performance of TSCP and alternative approaches on simulated data with $d = 2$ targets and heterogeneous Laplace noise. The target coverage level is 90%. Results are averaged over 200 trials. All methods achieve near-nominal coverage and approximate *Pop. Oracle* as the calibration sample size increases. TSCP tracks TSCP-LWC closely, with an almost indistinguishable average volume, highlighting the tightness of its rectangular enclosure.

The methods in Figure S9 approach the volume of *Pop. Oracle* as n increases. TSCP and TSCP-LWC have nearly indistinguishable mean volumes, suggesting that the rectangular enclosure adds little volume in this setting. TSCP-GWC is more conservative but produces smaller volumes than the data-splitting variant TSCP-S. Despite lacking a finite-sample coverage guarantee, the *Naive* approach performs well at larger calibration sizes, as also shown in Figure S10. Computational cost is examined separately below.

Figure S11 better highlights the prohibitive computational cost of TSCP-LWC, by reporting on the results of similar experiments where a small sample size is fixed, $|\mathcal{D}_{\text{cal}}| = 30$, while the outcome dimensions are gradually increased: $d \in \{2, 3, 4, 5, 10, 20, 30\}$.

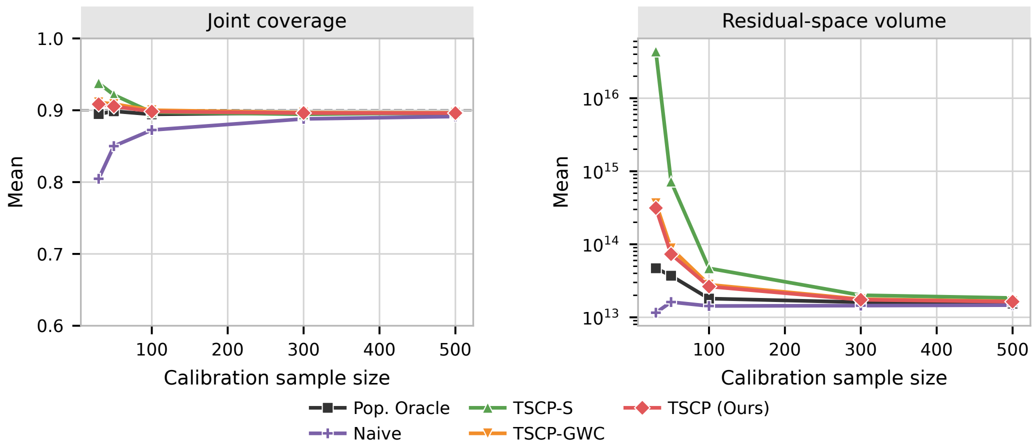


Figure S10: Performance of TSCP and alternative approaches on simulated data with $d = 10$ targets. Other details are as in Figure S9. TSCP-LWC is omitted from this figure due to its prohibitive cost with even moderately high-dimensional outcomes.

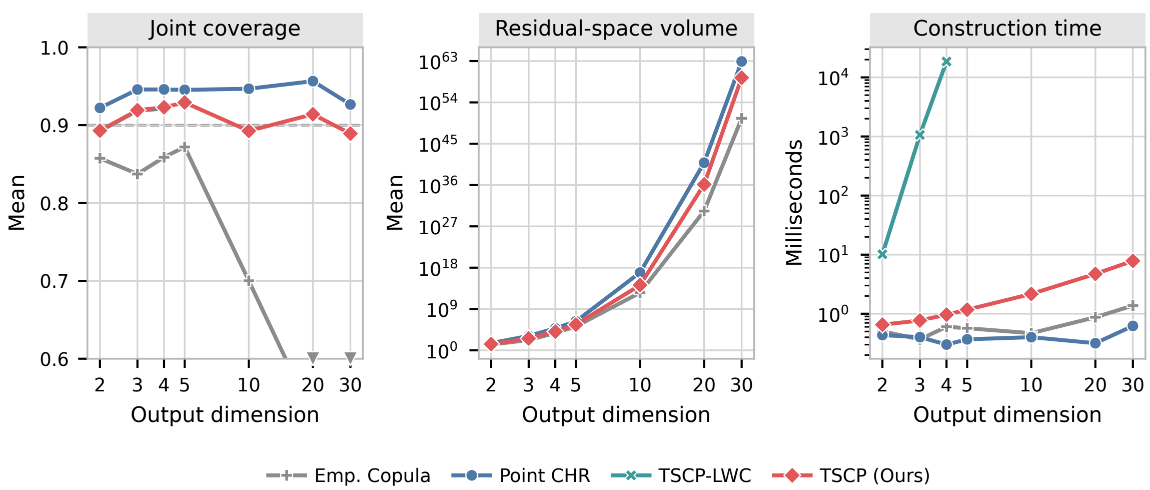


Figure S11: Absolute-residual experiment with Laplace noise, calibration size $n = 30$, and varying output dimension d. Results are averaged over 30 repetitions. The dashed line marks target coverage 0.9. TSCP-LWC is evaluated only at $d \in \{2, 3, 4\}$ because of its rapidly increasing computational cost.

C.10 Additional Real-Data Experiments

These diagnostics use 200 randomized splits per dataset with the same allocation and model settings as Section 4.4. All methods share the fitted model and observations within each split. The reported variation is over splits of each fixed dataset, not over independently sampled datasets from a population.

C.10.1 Coordinate-Wise Marginal Coverage and Interval Lengths

Figure S12 reports coordinate-wise marginal coverage for each of the six real datasets. The dashed line indicates the target joint coverage of 0.9, and each panel also reports TSCP's mean joint coverage. The coordinates follow the target-column order of each dataset, allowing for a clear comparison of how well each method performs across different dimensions of the outcome space.

To compare coordinate-wise lengths, Figure S13 reports the paired length ratio

$$A_{j,m} = \frac{1}{200} \sum_{r=1}^{200} \frac{\ell_{j,m}^{(r)}}{\ell_{j,\text{Unscaled}}^{(r)}}.$$

where $\ell_{j,m}^{(r)} = 2\mathcal{L}_{j,m}^{(r)}$ is the full outcome-space interval length for coordinate j, method m, and split r. Thus, a bar below one indicates a shorter interval on average relative to Unscaled Max on the same splits. Ratios are formed before averaging; they are not ratios of mean interval lengths.

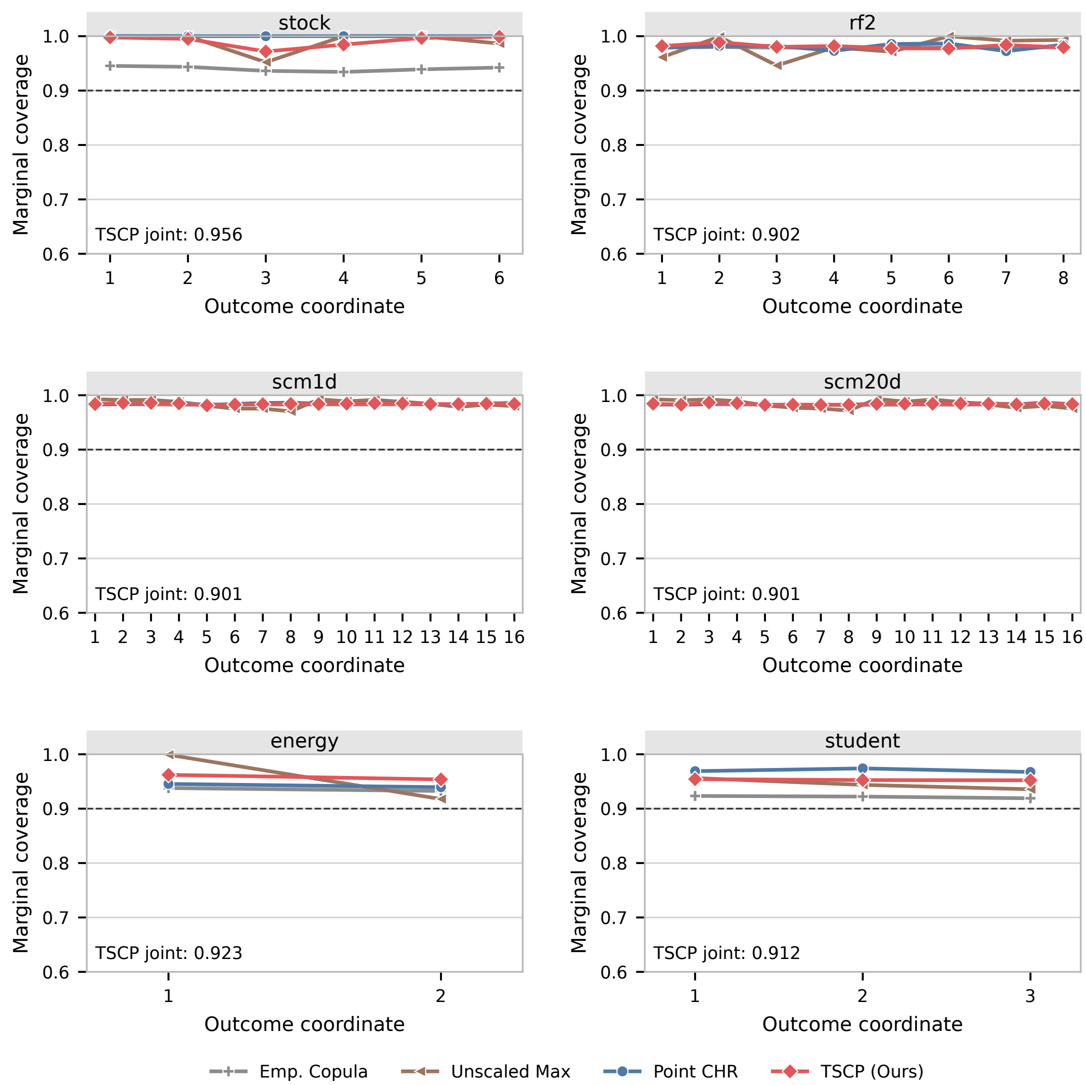


Figure S12: Coordinate-wise marginal coverage over 200 splits of each real dataset. The dashed line is the 0.9 joint target, and each panel also reports TSCP's mean joint coverage. Coordinates follow the target-column order of each dataset.

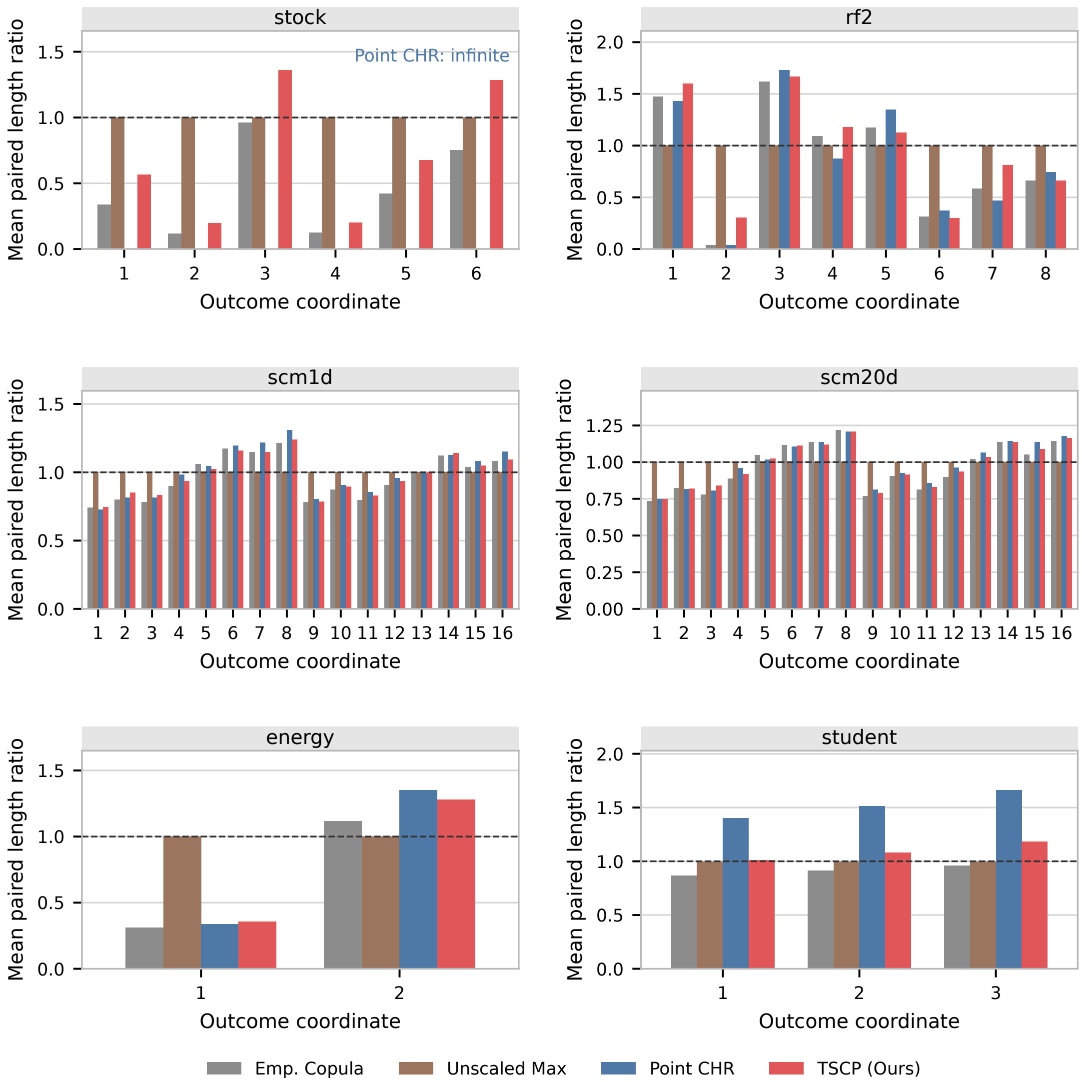


Figure S13: Coordinate-wise full interval lengths on all six real datasets, normalized within each split by the corresponding Unscaled Max length and then averaged over 200 splits. The dashed line marks a ratio of one. Point CHR is infinite on `stock`. These width comparisons should be read with the matching joint coverage and the marginal coverage in Figure S12.

C.10.2 Backward-Search and Fallback Diagnostics

We investigate the fallback and backward-search behavior of the TSCP implementation summarized in Algorithm 2. A fallback occurs when the mean cell has empty intersection with the global rectangle; a backward-search run occurs when that intersection is nonempty but the binary-search condition fails for at least one coordinate. We also record the number of coordinates using that branch and the number of candidate cells inspected before finding the boundary.

Table S8 reports the split-level and coordinate-level counts. Figure S14 shows the number of candidate cells inspected per coordinate and the construction time of the observed search regime. For each split, we time repeated TSCP calls on the same calibration residuals after a warm-up. The mean of the per-split median runtimes is

$$\frac{1}{200}\sum_{s=1}^{200} \mathrm{median}(T_{s1}, \dots, T_{sr}),$$

where T_{sj} is the runtime of the jth call on split s. We use $r = 7$ for $\alpha = 0.1$ and $r = 3$ for $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$. These calls are repeated timing measurements, not additional statistical repetitions. Regime-specific times average the medians only over splits exhibiting that regime. Timings exclude model fitting, residual computation, and diagnostic logging; calls run serially with native thread pools limited to one thread.

Dataset	n	Backward splits	Backward coordinates	Fallback splits	Binary only (ms)	Any backward (ms)	Fallback (ms)
`stock`	15	0/200	0/1200	0/200	1.007	–	–
`rf2`	384	0/200	0/1600	0/200	2.780	–	–
`scm1d`	490	0/200	0/3200	0/200	7.239	–	–
`scm20d`	448	0/200	0/3200	0/200	7.120	–	–
`energy`	38	0/200	0/400	0/200	0.466	–	–
`student`	32	0/200	0/600	0/200	0.620	–	–

Table S8: Executed TSCP search regimes at target coverage 0.9 on the six real datasets. Each split-count denominator is 200. The coordinate denominator is d times the number of non-fallback splits. Conditional times summarize the full TSCP call, not just the backward loop; a dash denotes an unobserved regime. n is the calibration size.

At the main target $\alpha = 0.1$, neither backward search nor fallback occurs in any of the 200 splits for any dataset. All 10,200 coordinate searches use the binary branch, with about 4.00–8.49 candidate-cell inspections per coordinate on average across datasets. Observing no events in these splits does not establish that backward search or fallback is impossible.

To explore further, we reuse the same 200 calibration residual matrices per dataset and vary the miscoverage level over $\alpha \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. In this experiment, we observe that backward search and fallback occur more frequently at higher miscoverage levels. On `energy` at $\alpha = 0.5$, backward search occurs in 3/200 splits (1.5%), while fallback occurs in 140/200 (70%). The remaining 57 splits use only binary search. At $\alpha = 0.7$ and 0.9, 10 energy splits use backward search and 190 fall back. In every observed backward-search case, each of the two coordinates inspects only *one* backward candidate. No backward search is observed in the other five datasets over the displayed levels. Fallback is often faster because

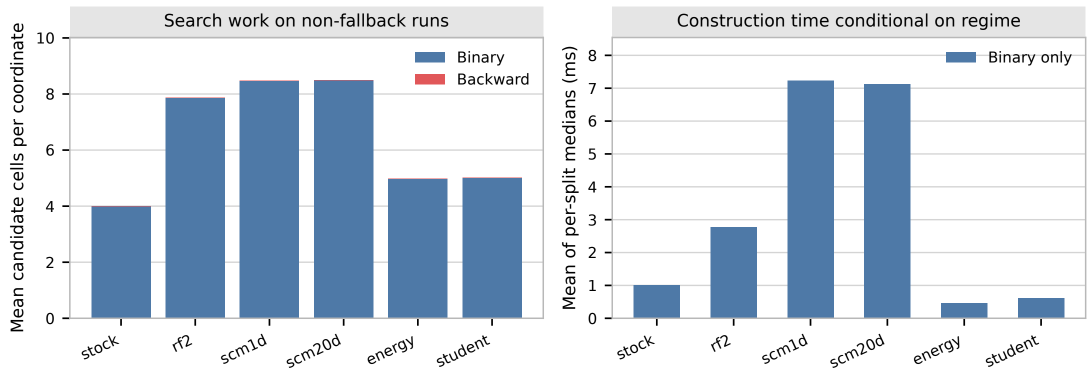


Figure S14: TSCP search work and construction time for 200 splits per real dataset at target coverage 0.9. Left: candidate cells inspected per coordinate, averaged over non-fallback runs. Counts include cells rejected by intersection checks. Right: construction times of the observed regime. Unobserved scenarios have no bars.

it returns the already computed global rectangle, so a lower average runtime at an extreme target need not indicate a more efficient local search.

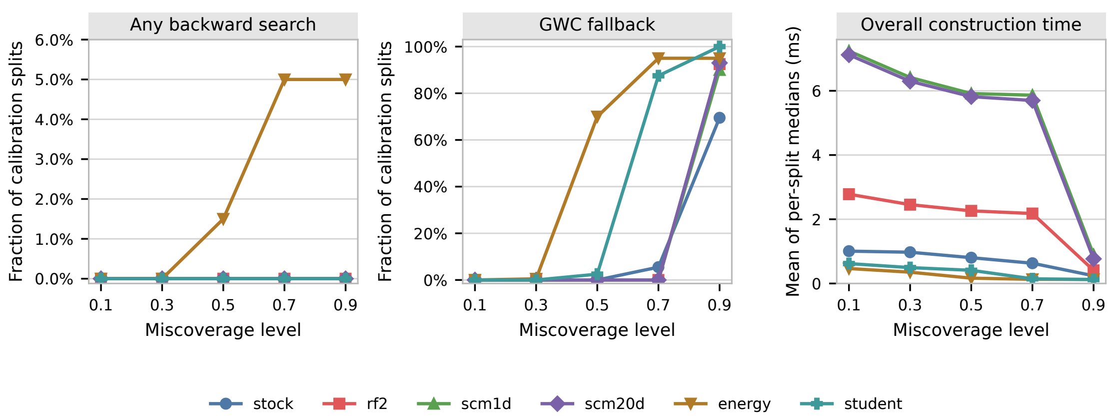


Figure S15: Computational stress test on the cached real-data calibration residuals. Each point summarizes 200 splits. The first two panels show the fractions of splits using any backward search or the global fallback. The final panel reports construction time.